Structural Change under Limited Growth Pass-Through*

BJOERN HARTMANN

University of St.Gallen

October 2026

Abstract

The pace of structural change towards services has slowed in recent decades, while aggregate markups have risen. This paper studies whether rising market power has contributed to this slowdown. I incorporate firms that differ in productivity and markups into a model of structural change. As firms adjust their markups in response to productivity growth, only part of these productivity gains is transmitted into lower sectoral prices. The strength of this *growth pass-through* is determined by the productivity distribution. Sectors with a thicker upper tail of the productivity distribution exhibit lower growth pass-through. I quantify the mechanism for the United States from 1980 to 2014, focusing on agriculture, industry, and services. I find that growth pass-through declined in all three sectors as markups rose. In a counterfactual with no market power, the expenditure share of services would have been about 3 percentage points higher by 2014, implying 31% more reallocation towards services over the sample period. The effect is primarily driven by larger price declines in industry. Because sectors are complements, this shifts expenditure away from industry and towards services.

Keywords: Structural Change; Markups; Growth Pass-Through; Firm Heterogeneity; Growth
JEL classification: O14; O41; L11; D43

*I am indebted to Reto Föllmi and Dominik Sachs for their continued guidance and support. I thank Franziska Bender, Marino Bucher, Francisco Buera, Ariel Burstein, Jonas Bruhin, Bruno Caprettini, Albert Flak, Costanza Gai, Agustín Gutiérrez, Rodolfo Manuelli, Marko Irisarri Martinez, Marc-Andreas Mündler, Julian Schärer, Davide Suverato, Florian Trouvain, and Ákos Valentinyi for valuable discussions and comments. I also thank participants at the SSES 2026 Meeting and seminar participants at Washington University in St. Louis, University of Manchester, and University of St. Gallen. I gratefully acknowledge the hospitality of Washington University in St. Louis and the University of Manchester as well as my hosts Francisco Buera and Akos Valentinyi during the writing of this chapter. All errors are my own. Email: bjoern.hartmann@unisg.ch

1 Introduction

Over recent decades, expenditure and employment have shifted away from goods production and towards services, but the pace of this reallocation has slowed. At the same time, market structure has changed markedly: large firms have expanded their market shares and the aggregate markup has risen. Standard models of structural change explain sectoral price movements through productivity growth, typically assuming that growth is fully transmitted into lower prices. These models therefore abstract from the role of changing markups in shaping relative prices. In this paper, I propose a parsimonious theory of structural change in which firms' endogenous markups weaken the transmission of productivity growth into sectoral prices. I use this framework to study how changes in market structure have contributed to the observed slowdown in structural change.

Markups affect structural change through two channels: they distort relative prices across sectors and, by raising the aggregate price level, lower real household income. Building on the demand framework of Comin et al. (2021), I introduce firms that differ in productivity and charge variable markups. The resulting incomplete transmission of productivity growth into sectoral prices affects both relative prices and the growth of real income. I quantify these channels for the United States 1980 to 2014, distinguishing three broad sectors: Agriculture, Industry, and Services. In a counterfactual with no market power, the expenditure share of Services would have been 3 percentage points higher, raising the expenditure share from 80% to 83% in 2014. Of the additional reallocation towards Services, at least 60% is explained by relative prices, while the remainder is driven by higher real income.

A central feature of my model is the transmission of productivity growth into sectoral prices, which I refer to as *growth pass-through*. I derive a simple closed-form expression for growth pass-through that depends solely on the productivity distribution. Sectors with a thicker upper tail exhibit lower growth pass-through, implying that identical productivity improvements can generate different price responses across sectors. As a result, observed sectoral price changes are no longer sufficient to identify underlying productivity growth. Instead, I use observed markups to discipline growth pass-through. In combination with sectoral price data, this allows me to recover underlying productivity growth.

The mechanism behind these results operates through the interaction of non-homothetic demand across sectors and variable-markup firms within sectors. On the demand side, consumers allocate expenditure across a set of sectoral composites using non-homothetic CES preferences. This upper-tier structure yields a decomposition of relative expenditure changes into a relative price effect and an income effect, thus allowing for a clean mapping from sectoral price indices

into expenditure shares.

Within each sector, I build on the framework of Melitz and Ottaviano (2008), with monopolistically competitive firms that differ in productivity and face quadratic demand. This demand structure implies a finite choke price and a variable price elasticity of demand at the variety level. As a result, markups are heterogeneous and increasing in firm productivity and size, while cost pass-through into prices is incomplete and lower for large firms. I show that introducing these features changes the mapping from productivity growth to structural change. The productivity distribution determines sectoral markups and growth pass-through, so that identical productivity growth generates differential price responses across sectors. This mechanism is absent under CES, where constant elasticities imply constant markups and complete growth pass-through.

Firm productivities are drawn from sector-specific Pareto distributions. Firms enter freely by paying a sunk entry cost. In equilibrium, only sufficiently productive firms actually produce, generating an endogenous productivity cut-off. The model preserves tractability and yields a simple closed-form characterisation of growth pass-through, depending solely on the Pareto shape parameter. Moreover, growth pass-through is given by the inverse of the cost-weighted sectoral markup. A thicker upper tail of the Pareto distribution therefore raises sectoral markups and lowers growth pass-through.

Growth pass-through is a sector-level equilibrium object, distinct from the firm-level pass-through studied in the empirical literature. Firm-level pass-through measures the response of an individual firm's price to marginal cost, holding the sectoral environment fixed. Growth pass-through instead captures the equilibrium response of the sectoral price index to a change in productivity. The latter incorporates the endogenous adjustment of prices and firm selection induced by entry.

Empirically, I combine US firm-level data from Compustat with sectoral expenditure and price data from the World Input-Output Database (WIOD). I aggregate the data into three broad sectors: Agriculture, Industry, and Services. By 2014, Agriculture exhibits the highest sectoral markup, followed by Industry and Services. I map these markups into the Pareto shape parameters using the relationship implied by the model. The resulting estimates imply a substantial decline in growth pass-through in all three sectors from 1980 to 2014: from about 90% to 63% in Agriculture, 89% to 79% in Industry, and 87% to 82% in Services. This has implications for the measurement of productivity growth. Conventional approaches that infer productivity from observed output and price movements abstract from the wedge created by variable markups. In my framework, sectoral price changes reflect both underlying productivity growth and its endogenous pass-through, so recovering technological change requires accounting for market structure as well. Therefore,

I combine the WIOD sectoral price changes with the growth pass-through estimates to recover sectoral productivity growth. In line with the literature, productivity growth is fastest in Agriculture and slowest in Services. On the demand side, estimates of the non-homothetic CES system imply gross complementarity across sectors and substantially different income elasticities, with Agriculture as relative necessity and Services as relative luxury goods.

I then calibrate the model to the observed evolution of US expenditure shares. The calibrated model closely tracks the long-run reallocation of expenditure away from Agriculture and Industry and towards Services. To quantify the contribution of limited growth pass-through, I construct a counterfactual that holds the recovered sectoral technology paths fixed while imposing complete growth pass-through. Under this counterfactual, the Services expenditure share in 2014 would have been 2.92 percentage points higher and the Industry share 2.67 percentage points lower than in the calibrated baseline. The additional increase in the Services share amounts to 31.4% of the structural change towards Services in the baseline model between 1980 and 2014 and corresponds to approximately 4.3 million additional workers in Services. About 63% of the additional reallocation from Industry towards Services is attributable to relative price effects, with the remainder due to income effects. A sectoral decomposition shows that the effect is driven primarily by Industry: imposing full pass-through in Industry alone raises the Services share by 3.20 percentage points, whereas imposing full pass-through only in Services lowers its own share by 0.63 percentage points. Because sectors are complements, larger price declines in Industry reduce its expenditure share and shift expenditure towards Services. Limited growth pass-through therefore slows structural change primarily by attenuating price declines in Industry.

I consider several extensions to assess the robustness and broader implications of the mechanism. First, replacing WIOD sectoral prices with BEA PCE price indexes, which provide an alternative price concept incorporating quality adjustment, leaves the large Industry to Services productivity gap intact and suggests that differential quality adjustment is unlikely to explain the main productivity pattern. Second, disaggregating the economy into eight sectors reveals substantial heterogeneity within Services: income elasticities, productivity growth, and growth pass-through differ markedly across Service industries, so structural change within Services need not be driven by a common mechanism. Finally, the baseline model can be extended to incorporate endogenous productivity growth driven by learning-by-imitation across firms. The firm productivity distribution then determines both the growth pass-through (as in the baseline) and the productivity growth rate in each sector. A thicker upper tail lowers growth pass-through, but also raises productivity growth. The two effects work in opposite directions: a sector may grow faster technologically while passing a smaller fraction of that growth into lower prices.

1.1 Related Literature

Structural Change. Structural change arises from two distinct mechanisms: income effects and price effects. Income effects arise from non-homothetic preferences, which shift expenditure shares from necessities towards luxuries as income rises (even when relative prices are constant). This is known as Engel's Law. The literature has proposed various preferences, including Stone-Geary (Buera and Kaboski, 2009; Herrendorf et al., 2013; Kongsamut et al., 2001), price independent generalised linearity (PIGL, see Boppart (2014); Herrendorf et al. (2021)), hierarchic preferences (Buera and Kaboski, 2012; Föllmi and Zweimüller, 2006, 2008; Matsuyama, 2002; Murphy et al., 1989), or non-homothetic CES (Auer et al., 2024; Baqaee and Burstein, 2023; Comin et al., 2021; Fally, 2022; Matsuyama, 2019; Redding and Weinstein, 2020).

Price effects are driven by asymmetric technologies across sectors, causing relative price changes. Baumol (1967) postulated productivity growth differences across sectors as driver of relative price changes. Ngai and Pissarides (2007) and Herrendorf et al. (2015) generate structural change using sectoral productivity growth. Acemoglu and Guerrieri (2008) employ sector-specific capital intensities and capital deepening to achieve price effects.

A growing body of literature combines both mechanisms. Boppart (2014) and Comin et al. (2021) perform a decomposition analysis to assess the relative contribution of income effects and price effects, and conclude that income effects are (at least weakly) more important. Herrendorf et al. (2013) show that the result depends on data: income effects dominate using final expenditure data, while price effects prevail with value-added data. Combinations of income and price effects have been applied to explain increases in the skill premium (Buera et al., 2022), structural change within the investment sector (Herrendorf et al., 2021), aggregate productivity differences across countries (Caselli, 2005; Duarte and Restuccia, 2010), or separating stagnant from progressive services (Duernecker et al., 2024). This paper contributes to this literature by splitting the price effect: productivity growth affects expenditure shares only to the extent that it is passed through into prices. With firm heterogeneity and variable demand elasticities, this pass-through is incomplete and becomes an additional determinant of structural change.

Markups. The measurement of markups using the production approach begins with Hall (1988) and advanced by Basu and Fernald (1997); Klette (1999); De Loecker and Warzynski (2012). The average markup has risen substantially since the 1980s. This is linked to various macroeconomic trends such as a falling labour share, increased market concentration and declining business dynamism (Autor et al., 2020; Barkai, 2020; De Loecker et al., 2020). Structural change itself is a driver of the rise in aggregate markup through compositional changes over time (Marto, 2024).

Beyond levels, the distribution of markups is relevant as well. More productive firms charge higher markups (Amiti et al., 2014, 2019; Berman et al., 2012; De Loecker and Warzynski, 2012; De Loecker et al., 2016, 2020), which can be linked to a firm's market power (Burstein et al., 2025). These firms also exhibit lower pass-through of cost-push shocks to prices (Gopinath and Rigobon, 2008; Gopinath, Itskhoki and Rigobon, 2010; Nakamura, 2008; Nakamura and Zerom, 2010; Campa and Goldberg, 2005). Dispersion in markups creates a wedge between the market equilibrium and the social optimum, which is a source of inefficiency. This lowers aggregate productivity, distorts the allocation of resources across firms and hinders firm entry (Baqaee and Farhi, 2020; Behrens et al., 2020; Burstein et al., 2024; Dhingra and Morrow, 2019; Edmond et al., 2023; Zhelobodko et al., 2012). In my theory, markups are pinned down by the price elasticity of demand. Under quadratic preferences, the price elasticity decline as firms become more productive. Because pass-through is incomplete, this (static) markup wedge has dynamic implications: as productivity evolves, the wedge continues to shape relative prices and hence expenditure shares.

Concentration. A large empirical literature documents rising market concentration across US industries (Kwon et al., 2024; Grullon et al., 2019), attributed variously to information-technology-driven increasing returns to scale (Lashkari et al., 2024), the growing importance of intangible capital (De Ridder, 2024), declining investment and business dynamism (Gutiérrez and Philippon, 2016), or a broader shift in the relationship between growth and rents (Aghion et al., 2023; Benkard et al., 2026). In my theory, the Pareto shape parameter plays the role of a concentration index: a thicker upper tail raises the cost-weighted markup. The extension in Section 6.3 allows sectoral markups to rise as technology advances, even with a fixed level of concentration. Rising concentration is therefore one driver of rising markups in this framework, but not the only one.

Technology Adoption. Building on Kortum (1997), who models innovation as an outcome of a research process, Alvarez, Buera and Lucas (2008, 2013) and Buera and Oberfield (2020) capture the idea that agents learn from surrounding agents through an exogenous process of meeting and learning. Lucas and Moll (2014) and Perla and Tonetti (2014) endogenise search effort by making firms actively search for better technology, at the cost of losing production while searching. Akcigit et al. (2018), Bloom, Schankerman and Van Reenen (2013) or Moretti (2021) provide compelling empirical evidence that spillovers and interactions among agents increase future productivity. The extension in Section 6.3 links this technology adoption to structural change by embedding learning-by-imitation into the multi-sector framework as a micro-foundation of the main model. Technology diffusion shifts the sectoral productivity distribution over time, and adoption affects expenditure

shares not only through growth, but also through its endogenous effect on markups and the pass-through of growth into prices.

The rest of the paper is structured as follows: In Section 2, I present the model and decompose the price effect into productivity growth and pass-through. In Section 3 describes the data, estimates cross-sector demand parameters, and measures markups. Section 4 calibrates the baseline model. Section 5, uses the calibrated model to quantify the effect of limited growth pass-through on structural change. Section 6 presents extensions to the model and the data. Section 7 concludes.

2 Theoretical Framework

Consider an economy with L homogeneous consumers, who each supply one unit of labour inelastically. They spend all their income on consumption. Labour is the only factor of production, it is perfectly mobile across sectors, and the labour market is perfectly competitive. There are finitely many sectors $\mathcal{J}$, each with a continuum of differentiated varieties. Consumption varieties are supplied by monopolistically competitive firms. Each firm produces exactly one variety under constant returns to scale. The wage rate serves as numéraire. The model characterises the static equilibrium for a given set of sectoral technology parameters. In the quantitative analysis, I apply this static equilibrium repeatedly to the observed evolution of these parameters over time.

This section is structured as follows: 2.1 specifies preferences: an upper-tier non-homothetic CES over sectors and a lower-tier quadratic utility over varieties. 2.2 introduces heterogeneous monopolistic-competition firms and derives pricing. 2.3 characterises the within-sector equilibrium. 2.4 analyses the main results of the equilibrium.

2.1 Consumers and Preferences

2.1.1 Upper-Tier Utility

Consumers have preferences across sectors $\mathcal{J}$ represented by a non-homothetic CES aggregator defined over sectoral composites. Let Q denote the aggregate real consumption index per consumer which aggregates the sectoral consumption bundles $\{Q_j\}_{j=1}^{J}$. Preferences across sectors are implicitly defined by the constraint

$$\sum_{j\in\mathcal{J}} \beta_j^{\frac{1}{\eta}} Q_j^{\frac{\eta-1}{\eta}} Q^{\frac{\epsilon_j-\eta}{\eta}} = 1, \qquad 0<\eta<1, \tag{1}$$

where $\beta_j > 0$ describes the sector-specific demand shifter (with $\sum_{j=1}^{J} \beta_j = 1$), η is the elasticity of substitution across sectors and $\epsilon_j > 0$ governs the income elasticity of demand of sector j.[1]

Let E denote total expenditure per consumer and let E_j denote expenditure on sector j. We define a sectoral deflator P_j and a corresponding real composite Q_j such that

$$E_j \equiv P_j Q_j, \qquad \omega_j \equiv \frac{E_j}{E}, \qquad \sum_{j \in \mathcal{J}} \omega_j = 1. \tag{2}$$

where we interpret ω_j as the expenditure share of sector j. We further define an aggregate deflator P by $E \equiv PQ$.[2] Under (1) and (2), the implied demand system yields

$$Q_j = \beta_j \left(\frac{P_j}{P} \right)^{-\eta} Q^{\epsilon_j}, \tag{3}$$

and the associated aggregate price index:

$$P = \left(\sum_{j \in \mathcal{J}} \beta_j P_j^{1-\eta} Q^{(\epsilon_j - 1)} \right)^{\frac{1}{1-\eta}}. \tag{4}$$

Combining (3) with (1), we get the expenditure share of sector j:

$$\omega_j = \beta_j \left(\frac{P_j}{P} \right)^{1-\eta} Q^{(\epsilon_j - 1)} \tag{5}$$

This expression illustrates the effect of the sectoral income elasticity ϵ_j on spending patterns. *Ceteris paribus*, an increase in aggregate consumption will result in a larger spending share of sector j if and only if $\epsilon_j > 1$, and in a smaller spending share if $\epsilon_j < 1$. See Appendix A.1 for details.

With the wage normalised to one, each consumer earns labour income $E = 1$, so aggregate

[1] The preferences described here are based on the work of Comin et al. (2021). They have subsequently been applied in various settings, e.g. in Auer et al. (2024); Baqaee and Burstein (2023); Fally (2022); Matsuyama (2019). This utility specification nests several classical utility functions. Setting $\epsilon_j = 1 \; \forall j$ yields the standard CES aggregator. Additionally letting $\eta \to 1$ corresponds to Cobb-Douglas preferences. Note that the sector-specific demand-shifters β_j could be time-dependent, which would capture taste shocks to preferences, as applied in Baqaee and Burstein (2023) and Redding and Weinstein (2020) using the nonhomothetic CES preferences.

[2] With non-homothetic within-sector preferences (as in the quadratic lower tier introduced below), the welfare-relevant sectoral expenditure function takes the form $E_j = e_j(\mathbf{p}_j, U_j)$ and is generally not proportional to the sectoral utility level. Hence, the scalar P_j in (2) should be interpreted as a *measurement* deflator (a unit-value index) associated with the sectoral composite Q_j, rather than as a welfare-consistent cost-of-living index. The closed-form demand system (3)–(5) therefore serves as a tractable representation of expenditure allocation over *measured* sectoral aggregates. Welfare analysis would require a separate money-metric construction based on the sectoral indirect utility or expenditure function.

expenditure equals L. Since labour is the only factor of production, aggregate sectoral expenditure equals sectoral labour income:

$$L_j = LE_j = L\omega_j.$$

Properties of Demand System. Having established consumer demand at the sectoral level, we can analyse the properties of this demand system. First, holding aggregate real consumption constant, the elasticity of substitution between goods of different sectors is constant and given by:

$$\frac{\partial \log(Q_j/Q_k)}{\partial \log(P_k/P_j)} = \eta, \qquad \forall\, j,k \in \mathcal{J},\ j \neq k$$

This property is identical with the elasticity of substitution in CES preferences. Second, holding relative prices constant, the elasticity of relative demand (and of relative expenditure) with respect to real consumption is also constant and given by:

$$\frac{\partial \log(Q_j/Q_k)}{\partial \log Q} = \frac{\partial \log(\omega_j/\omega_k)}{\partial \log Q} = (\epsilon_j - \epsilon_k), \qquad \forall\, j,k \in \mathcal{J},\ j \neq k$$

This elasticity depends only on the sectoral income elasticities ϵ_j. Suppose that $\epsilon_j < \epsilon_k$, then, as the economy grows, consumption is reallocated from sector j to k, because k is a *relative* luxury good.[3] Hence, the Engel curves are non-linear and the preferences are nonhomothetic. Importantly, this effect does not vanish as income grows (as would be the case with Stone-Geary).

Both properties jointly imply that it is possible to fully separate price effects and income effects when measuring sectoral expenditure shares.[4] Relative consumption expenditure can be expressed as:

$$\log\left(\frac{\omega_j}{\omega_k}\right) = \underbrace{(1-\eta)\log\left(\frac{P_j}{P_k}\right)}_{\text{Price Effect}} + \underbrace{(\epsilon_j - \epsilon_k)\log Q}_{\text{Income Effect}} + \underbrace{\log\left(\frac{\beta_j}{\beta_k}\right)}_{\text{Constant}} \qquad \forall\, j,k \in \mathcal{J},\ j \neq k \quad (6)$$

[3] A similar feature can be achieved with a hierarchy of needs, where consumers start buying new products with high income elasticity as they become richer, while older luxury products become necessity goods over time. Such hierarchic preferences have been studied by e.g. Murphy, Shleifer and Vishny (1989), Föllmi and Zweimüller (2006, 2008) or Buera and Kaboski (2012).

[4] This complete separation of income and price effects is possible because the nonhomothetic CES preferences are *implicitly additively separable*. *Explicitly additively separable* preferences (such as Stone-Geary (e.g. Kongsamut et al. (2001), Herrendorf et al. (2013), Buera and Kaboski (2009)), price independent generalised linearity (PIGL, see Boppart (2014); Herrendorf et al. (2021)), constant ratio of income elasticity (CRIE, see Caron et al. (2014))) do not allow for this clear separation. See Matsuyama (2019) for an excellent discussion on the distinction between explicit and implicit additivity.

The first term captures price effects, which diminish as $\eta \to 1$. The easier it is for consumers to shift consumption across sectors, the less price differences matter for expenditure shares. The second term encompasses income effects, which are purely determined by income elasticities ϵ_j. As consumers become richer, they shift spending towards the *relative* luxury good. The last term is a constant and captures tastes / demand-shifters.

2.1.2 Lower-Tier Utility

Conditional on sectoral expenditure E_j chosen at the upper tier, consumers allocate spending across differentiated varieties within sector j according to a quadratic sub-utility. Let Ω_j denote the set of available varieties in sector j. Consumers have preferences over a continuum of differentiated varieties that are represented by the quadratic sub-utility

$$U_j = \int_{i \in \Omega_j} \left(\alpha q_{ij} - \frac{\gamma}{2} q_{ij}^2 \right) \, di, \qquad \alpha, \gamma > 0 \tag{7}$$

where q_{ij} is individual demand for variety i in sector j, and α and γ are demand parameters, assumed to be common across sectors. Quadratic utility is characterised by a variable elasticity of substitution across varieties and a finite marginal utility at zero consumption, i.e. a firm faces a finite choke price above which demand for its variety is zero.[5] Combined with firm heterogeneity, this specification provides a parsimonious way to generate heterogeneous markups and incomplete firm pass-through while preserving analytical tractability and admitting closed-form solutions.

Individuals take prices as given and maximise (7) subject to their sectoral budget constraint:

$$\int_{i \in \Omega_j} p_{ij} q_{ij} \, di = E_j \tag{8}$$

where p_{ij} is the price of variety i in sector j and E_j is expenditure in that sector.

Individual Demand. Each consumer's demand for variety i in sector j follows from maximising

[5] Specifically, the elasticity of substitution between varieties within a sector is given by: $\sigma_j(q) = \frac{d \ln(q_{hj}/q_{ij})}{d \ln MRS_{ihj}}|_{q_{ij}=q_{hj}} = -\frac{u_j'(q)}{q u_j''(q)}$. Using the demand function implied by quadratic preferences, this can also be expressed as: $\sigma_j(q) = \frac{p_{ij}}{\bar{p}_j - p_{ij}}$, which is not constant and depends on prices. The marginal utility at zero consumption is given by α. More generally, quadratic preferences imply a subconvex demand curve (see Mrázová and Neary, 2017), thus fulfilling Marshall's Second Law of Demand, and belong to the class of preferences introduced by Pollak (1971). See, *inter alia*, Dhingra (2013); Dixit (1979); Eckel and Neary (2010); Mayer et al. (2014); Melitz and Ottaviano (2008); Parenti et al. (2017); Ottaviano and Suverato (2024) for applications of quadratic utility in various settings.

the quadratic sub-utility (7) subject to the budget constraint (8):

$$q_{ij} = \begin{cases} \dfrac{\alpha}{\gamma}\left(1 - \dfrac{p_{ij}}{\bar{p}_j}\right), & p_{ij} \leq \bar{p}_j, \\ 0, & p_{ij} > \bar{p}_j, \end{cases} \tag{9}$$

where $\bar{p}_j = \alpha/\lambda_j$, with λ_j the Lagrange multiplier on the sectoral budget constraint. The price $\bar{p}_j$ is the sectoral choke price: demand for a variety is zero when its price reaches $\bar{p}_j$ and remains zero for any higher price. For varieties with positive demand, demand is linear in its own price. The relative price (meaning relative to the sectoral choke price) is a sufficient statistic for demand. When the price is zero, demand simplifies to α/γ.

Price Elasticity of Demand. For varieties with strictly positive demand ($p_{ij} < \bar{p}_j$), the price elasticity of demand for variety i in sector j, holding the sectoral choke price fixed, is defined as:

$$\varepsilon_{ij} = -\frac{d \log q_{ij}}{d \log p_{ij}}.$$

Given the demand function (9), and defining $x_{ij} = p_{ij}/\bar{p}_j$, we have:

$$\varepsilon_{ij} = \frac{p_{ij}/\bar{p}_j}{1 - p_{ij}/\bar{p}_j} = \frac{x_{ij}}{1 - x_{ij}}. \tag{10}$$

The elasticity is increasing in the variety's own price p_{ij}, implying that consumers become less price sensitive as prices fall (or quantities increase), which is known as Marshall's Second Law of Demand. As shown later, the relevant domain is $x_{ij} \in (0.5, 1)$.

2.2 Firms and Production

Firm entry is unrestricted but costly. Each potential producer in sector j must pay an exogenous sunk cost $F > 0$ (in units of labour) to develop a new technology. After paying this cost, the firm learns its idiosyncratic productivity $a_{ij} > 0$, drawn from a continuous Pareto distribution with CDF

$$G_j(a) = 1 - \left(\frac{A_j}{a}\right)^{\kappa_j}, \qquad a \geq A_j,$$

where A_j is the sector-specific lower bound of the productivity distribution and $\kappa_j > 1$ its shape parameter. After observing its productivity, the firm produces only if it can generate positive demand; otherwise it exits without producing. Production uses labour as the only input. A firm with

productivity a_{ij} produces according to $y_{ij} = a_{ij} l_{ij}$, where l_{ij} denotes the firm's labour input.

Pricing. Given its productivity and the demand function (9), each firm chooses its price to maximise operating profits. Firm i in sector j solves

$$\max_{p_{ij} \leq \bar{p}_j} \pi_{ij} = \left(p_{ij} - \frac{1}{a_{ij}} \right) y_{ij}, \qquad y_{ij} \equiv L q_{ij},$$

taking the choke price $\bar{p}_j$ as given. The first-order condition yields the pricing rule:

$$p_{ij} = \frac{1}{2} \left(\bar{p}_j + \frac{1}{a_{ij}} \right)$$

The optimal price is the arithmetic average between the firm's marginal cost $1/a_{ij}$ and the sectoral demand ceiling $\bar{p}_j$. For firms with $1/a_{ij} < \bar{p}_j$, this implies that the relative price

$$x_{ij} \equiv \frac{p_{ij}}{\bar{p}_j} = \frac{1}{2} \left(1 + \frac{1}{a_{ij} \bar{p}_j} \right)$$

lies in the interval $x_{ij} \in (0.5, 1)$: the relative price of any viable producer cannot be less than one half. Firms with higher productivity set lower prices than low-productivity firms. Let a_j^* denote the productivity of the marginal producer (who is indifferent between producing and not producing), defined by $p_{ij} = \bar{p}_j$. Evaluating the pricing rule at $a_{ij} = a_j^*$ and imposing $p_{ij} = \bar{p}_j$ yields $\bar{p}_j = 1/a_j^*$. Recall that $\bar{p}_j = \alpha / \lambda_j$. Combining both expressions pins down the Lagrange multiplier at $\lambda_j = \alpha a_j^*$.

Firms with productivity $a_{ij} < a_j^*$ do not produce in equilibrium, because their productivity is too low to generate positive demand. Since the choke price equals the inverse of the productivity cut-off, we can rewrite the pricing rule in terms of the cut-off level:

$$p_{ij} = \frac{1}{2} \left(\frac{1}{a_{ij}} + \frac{1}{a_j^*} \right) \tag{11}$$

Markups & Firm Pass-through. The relative price x_{ij} is a sufficient statistic to determine a firm's markup and pass-through (Arkolakis and Morlacco, 2017). The markup is fully determined by the price elasticity of demand and given by: [6]

[6] To see this, set up a generic profit maximisation problem of a firm. Profits are given by $\pi_{ij} = \left(p_{ij} - a_{ij}^{-1} \right) q_{ij}$, where a_{ij} denotes firm productivity (so marginal cost is $c_{ij} = 1/a_{ij}$). Profit maximisation yields $q_{ij} + p_{ij} \frac{dq_{ij}}{dp_{ij}} - \frac{1}{a_{ij}} \frac{dq_{ij}}{dp_{ij}} = 0$. Define the markup as price over marginal cost. The first-order condition can be rearranged to $\mu_{ij} =$

$$\mu_{ij} = \frac{\varepsilon_{ij}}{\varepsilon_{ij} - 1} = \frac{1}{2 - 1/x_{ij}} = \frac{1}{2}\left(1 + \frac{a_{ij}}{a_j^*}\right) \geq 1 \tag{12}$$

The markup decreases in the relative price. The closer a firm's price to the choke price, the lower its markup. Conversely, high-productivity firms will charge higher markups (Amiti et al., 2014, 2019; Berman et al., 2012; De Loecker and Warzynski, 2012; De Loecker et al., 2016, 2020). Coupled with lower prices, this implies a positive relationship between sales and markups. This result is driven by the variable price elasticity of demand: as quantities increase, consumers become less price sensitive. A high-sales firm exploits this willingness-to-pay by increasing its markup.

Firm-level pass-through measures the elasticity of a firm's price with respect to marginal cost, holding the sectoral choke price fixed. Let $c_{ij} = 1/a_{ij}$ denote marginal cost. Using the pricing rule in (11), we get:

$$\Phi_{ij} \equiv \left.\frac{\partial \log p_{ij}}{\partial \log c_{ij}}\right|_{\bar{p}_j} = \frac{a_j^*}{a_{ij} + a_j^*} = \frac{2x_{ij} - 1}{2x_{ij}} \in (0, 1/2) \tag{13}$$

Firm-level pass-through is therefore incomplete and decreases with productivity. Large, high-productivity firms charge higher markups and absorb a greater fraction of marginal-cost changes through markup adjustments, while firms close to the productivity cut-off exhibit higher pass-through (Campa and Goldberg, 2005; Gopinath and Rigobon, 2008; Gopinath et al., 2010; Nakamura, 2008; Nakamura and Zerom, 2010). The intuition is simple: low-productivity firms barely have any markup to begin with. Hence, a larger share of marginal-cost changes must be reflected in prices.[7] See Appendix A.1 for a detailed derivation of these results. Figure 1 summarises the key implications of the quadratic utility on firm behaviour. It illustrates the relationship between prices, markups, and pass-through, normalising the productivity cut-off a_j^* to one for simplicity. The blue line shows how firm prices decline with productivity according to (11), while the orange line depicts the increasing markup implied by (12). The black line shows that firm-level pass-through decreases with productivity, as more productive firms absorb a larger fraction of marginal-cost changes through markup adjustments.

$\frac{1}{1+\frac{dp_{ij}}{dq_{ij}}\frac{q_{ij}}{p_{ij}}} = \frac{\varepsilon_{ij}}{\varepsilon_{ij}-1}$, which implies the profit-maximising price $p_{ij} = \mu_{ij}c_{ij} = \mu_{ij}/a_{ij}$.

[7] A comparison to standard CES preferences may be in order. The elasticity of substitution under CES is constant (hence the name) and given by σ. This implies constant markups for all firms, irrespective of their production possibilities. This results in constant and complete pass-through of cost-push shocks to prices ($\Phi = 1$).

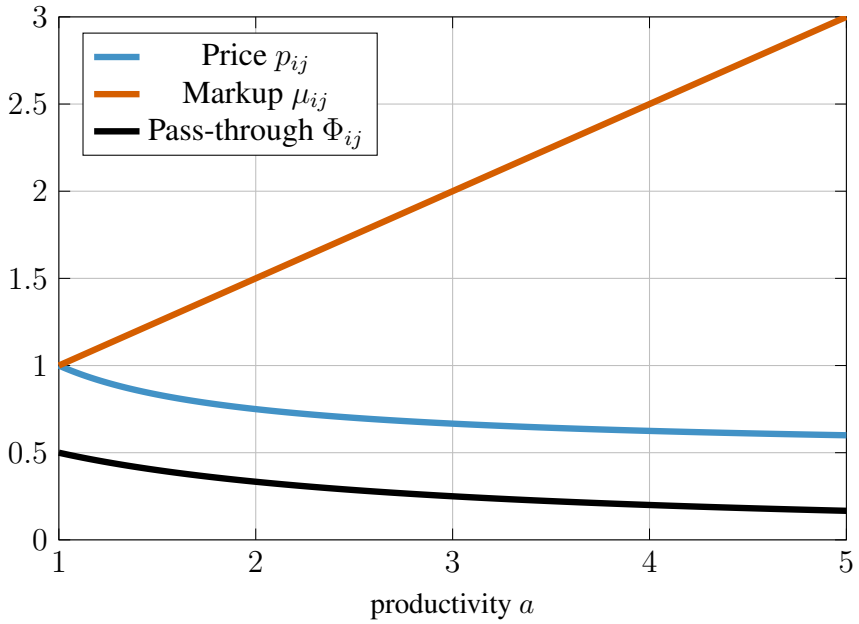


Figure 1: Firm Prices, Markups, and Pass-through under Quadratic Demand

Firm Output & Profits. With prices, productivity and demand established, we can determine total firm demand and profits. Since consumers are homogenous, aggregate firm demand is given by: $y_{ij} = Lq_{ij}$. Using Marshallian demand (9) and prices (11), aggregate demand and profits are:

$$y_{ij} = L\left(\frac{\alpha}{2\gamma}\right)\left(1 - \frac{a_j^*}{a_{ij}}\right) \tag{14}$$

$$\pi_{ij} = (p_{ij} - 1/a_{ij})y_{ij} = L\left(\frac{\alpha}{4\gamma a_j^*}\right)\left(1 - \frac{a_j^*}{a_{ij}}\right)^2 \tag{15}$$

Firms with higher productivity enjoy both higher quantities and higher markups, and therefore earn higher profits.

2.3 Equilibrium

Under the assumption of free entry, the equilibrium in each sector j is determined by a free-entry condition. Expected operating profits prior to observing the productivity draw equal the sunk cost of entry. Formally, we have:

$$\int_{a_j^*}^{\infty} \pi_j(a) dG_j(a) = F \qquad \forall j \in \mathcal{J} \tag{FE}$$

The combination of quadratic demand, linear production technology, and Pareto distributed firm productivity yields a closed-form expression for the equilibrium productivity cut-off:

$$a_j^* = \left(\frac{L\delta_\pi(\kappa_j)}{F}\right)^{\frac{1}{1+\kappa_j}} A_j^{\frac{\kappa_j}{1+\kappa_j}}, \qquad \delta_\pi(\kappa_j) = \left(1 - \frac{2\kappa_j}{\kappa_j+1} + \frac{\kappa_j}{\kappa_j+2}\right)\frac{\alpha}{4\gamma} \tag{16}$$

A high value of κ_j lowers the cut-off value. As $\kappa_j \to \infty$, we have $a_j^* \to A_j$. The reason is that a high value of κ_j implies the existence of many low-productivity firms, so it is unlikely to get a good draw upon paying the fixed costs. In order to break even in equilibrium, firms require a lower cut-off level a_j^*. Hence, high κ_j leads to little selection and a greater concentration of firms near the productivity cut-off, where firm-level pass-through is higher. As shown below, a higher κ_j also implies stronger growth pass-through. The cut-off decreases in the fixed costs F and rises in the market size L.

Labour market (and goods market) clearing imply that the aggregate sales of producing firms equal expenditure in sector j:

$$N_j \int_{a_j^*}^{\infty} p_j(a) y_j(a) \frac{dG_j(a)}{1 - G_j(a_j^*)} = L_j \tag{LMC}$$

where N_j is the number of producing firms in sector j and $y_j(a)$ denotes aggregate demand of a variety with productivity a in sector j. Solving (LMC) for N_j yields:

$$N_j = \frac{L_j}{F}\frac{\delta_\pi(\kappa_j)}{\delta_x(\kappa_j)}[1 - G_j(a_j^*)] = \frac{L_j}{F}\frac{1}{1+\kappa_j}\left(\frac{A_j}{a_j^*}\right)^{\kappa_j}, \qquad \delta_x(\kappa_j) = \left(1 - \frac{\kappa_j}{\kappa_j+2}\right)\frac{\alpha}{4\gamma} \tag{17}$$

Finally, we can determine prices and real consumption. Define a real quantity aggregator Q_j over consumed varieties:

$$Q_j = \int_0^{N_j} q_{ij} \; di$$

We define P_j as the corresponding unit-value deflator, $P_j \equiv E_j/Q_j$, so that $E_j = P_j Q_j$ holds by construction. Per capita expenditure in sector j, the sectoral quantity index and the sectoral price

index are given by:

$$E_j = \int_0^{N_j} p_{ij} q_{ij}\, di = \frac{\alpha N_j}{2\gamma a_j^*} \frac{1}{\kappa_j + 2} \tag{18}$$

$$Q_j = \int_0^{N_j} q_{ij} di = \frac{\alpha N_j}{2\gamma} \frac{1}{\kappa_j + 1} \tag{19}$$

$$P_j = E_j / Q_j = \frac{(\kappa_j + 1)}{(\kappa_j + 2)} \frac{1}{a_j^*} \tag{20}$$

See Appendix A.2 for a derivation. The sectoral price index P_j is inversely proportional to the sectoral productivity cut-off a_j^*, so any force that raises a_j^* (e.g. larger market size L or lower fixed costs F) reduces the sectoral price index and, for a given E_j, raises real consumption E_j/P_j. A thicker upper tail of the sectoral productivity distribution (low κ_j) implies tougher selection, more high-productivity firms, a higher a_j^*, and thus a lower price index P_j, whereas a thinner tail (high κ_j) weakens selection and increases P_j. Equations (16)–(20) highlight how sectors differ in equilibrium when the Pareto parameters (κ_j, A_j) are heterogeneous. Sectors with a high lower bound A_j (good underlying technology) or a thick upper tail (low κ_j, many potential low-cost firms) have a high cut-off a_j^* and thus a low sectoral price index P_j. With non-homothetic CES preferences, sectors differ not only in prices P_j but also in income elasticities ϵ_j. A low sectoral price index P_j lowers the expenditure share through the standard substitution channel (governed by $\eta < 1$), while a high income elasticity $\epsilon_j > 1$ makes the sector expand as real consumption Q grows. Sectoral productivity differences thus translate into systematic differences in prices, employment, and the number of firms across sectors.

2.4 Analytical Results

2.4.1 Aggregate Markup

Define the sector-level aggregate markup as the cost-weighted average of firm-level markups. In sector j, a firm with productivity a has marginal cost $c(a) = a^{-1}$, price $p(a)$, and markup $\mu(a) \equiv p(a)c(a)^{-1} = ap(a)$. Its variable cost equals $c(a)y(a) = a^{-1}y(a)$. The cost-weighted sectoral markup is therefore

$$\mu_j^c \equiv \frac{\int_{a_j^*}^{\infty} \mu(a)\, a^{-1} y(a)\, dG_j(a)}{\int_{a_j^*}^{\infty} a^{-1} y(a)\, dG_j(a)} = \frac{\int_{a_j^*}^{\infty} p(a)\, y(a)\, dG_j(a)}{\int_{a_j^*}^{\infty} a^{-1} y(a)\, dG_j(a)},$$

The second equality uses $\mu(a)c(a) = p(a)$. Hence μ_j^c can be written as the ratio of sectoral revenue to sectoral variable cost. Substituting the equilibrium price and quantity schedules, and using the Pareto distribution yields:

$$\mu_j^c = 1 + \frac{1}{\kappa_j} \tag{21}$$

The distortion in sector j depends only on the shape parameter κ_j. Sectors with a thick upper tail have higher cost-weighted markups. This mechanism is consistent with Burstein et al. (2025), who document that sector-level markups rise with concentration.

At the aggregate level, define sectoral variable cost as

$$C_j = N_j \int_{a_j^*}^{\infty} a^{-1} y(a) \frac{dG_j(a)}{1 - G_j(a_j^*)}$$

and let $C = \sum_j C_j$ denote total variable cost in the economy. Aggregate sectoral expenditure is LE_j, and with the wage normalised to one, this equals sectoral labour income L_j, so $LE_j = L_j$. Since the cost-weighted markup satisfies $\mu_j^c = (LE_j)/C_j$ and $\mu_j^c = 1 + 1/\kappa_j$, it follows that variable cost in sector j is a constant fraction of sectoral employment,

$$C_j = \frac{LE_j}{\mu_j^c} = \frac{L_j}{\mu_j^c} = \frac{\kappa_j}{\kappa_j + 1} L_j.$$

Then, the aggregate markup is given by:

$$\mathcal{M} \equiv \frac{\sum_j \mu_j^c C_j}{\sum_j C_j} = \sum_j \mu_j^c \, \omega_j^c, \qquad \omega_j^c \equiv \frac{C_j}{C} = \frac{\frac{\kappa_j}{\kappa_j + 1} L_j}{\sum_{m \in \mathcal{J}} \frac{\kappa_m}{\kappa_m + 1} L_m}, \tag{22}$$

where ω_j^c is the variable cost share of sector j, such that $\mathcal{M}$ is a cost-weighted average of sectoral markups.[8] With homogeneous technology shapes ($\kappa_j = \kappa$ for all j), the aggregate markup is constant at $\mathcal{M} = 1 + 1/\kappa$. With heterogeneous κ_j, sectors with a thicker upper tail (and hence higher μ_j^c) contribute more to the aggregate distortion. As structural change reallocates employment toward sector j, its labour share—and hence its variable-cost share ω_j^c—naturally rises, so the sectoral markup μ_j^c receives a larger weight in the aggregate markup $\mathcal{M}$. When the shape parameters κ_j differ across sectors, this changes the aggregate markup over time, making struc-

[8] Note that the variable cost share ω_j^c generally does not equal the expenditure share ω_j. They are identical if and only if $\kappa_j = \kappa \quad \forall j \in \mathcal{J}$. To see this, write $\omega_j^c = \frac{C_j}{C} = \frac{L_j/\mu_j^c}{L/\mathcal{M}}$. When κ is identical across sectors, this can be simplified to $\frac{L_j/\mu}{L/\mu} = \frac{L_j}{L} = \omega_j = E_j$.

tural change itself a driver of economy-wide markup dynamics, as studied by Marto (2024). See Appendix A.3 for a derivation of (21) and the sales-weighted markup as comparison.

2.4.2 Growth Pass-Through

Now, having established the equilibrium allocation across sectors, we can turn to growth. I define (exogenous) productivity growth as an increase in the lower bound of the productivity distribution A_j. The exogenous shift in A_j is the heterogeneous-firms analogue of a sector-specific TFP shock in a representative firm model. This shifts the distribution of potential entrants and raises the equilibrium productivity cut-off a_j^*, which in turn lowers the sectoral price index. The response of the sectoral price index can be summarised by the following proposition:

Proposition 1. *Consider an equilibrium described above (16)–(20) with non-homothetic CES preferences across sectors, $0 < \eta < 1$, and quadratic demand within sectors. Let A_j denote the lower bound of the productivity distribution in sector j, $\kappa_j > 1$ the shape parameter, and P_j the sectoral price index. A sector-specific productivity change in sector j, modelled as a change in the lower bound A_j, affects the sectoral price index according to*

$$\rho_j \equiv -\frac{d\log P_j}{d\log A_j} = -\frac{d\log P_j}{d\log a_j^*}\frac{d\log a_j^*}{d\log A_j} = \frac{\kappa_j}{1+\kappa_j} \in (1/2, 1) \tag{23}$$

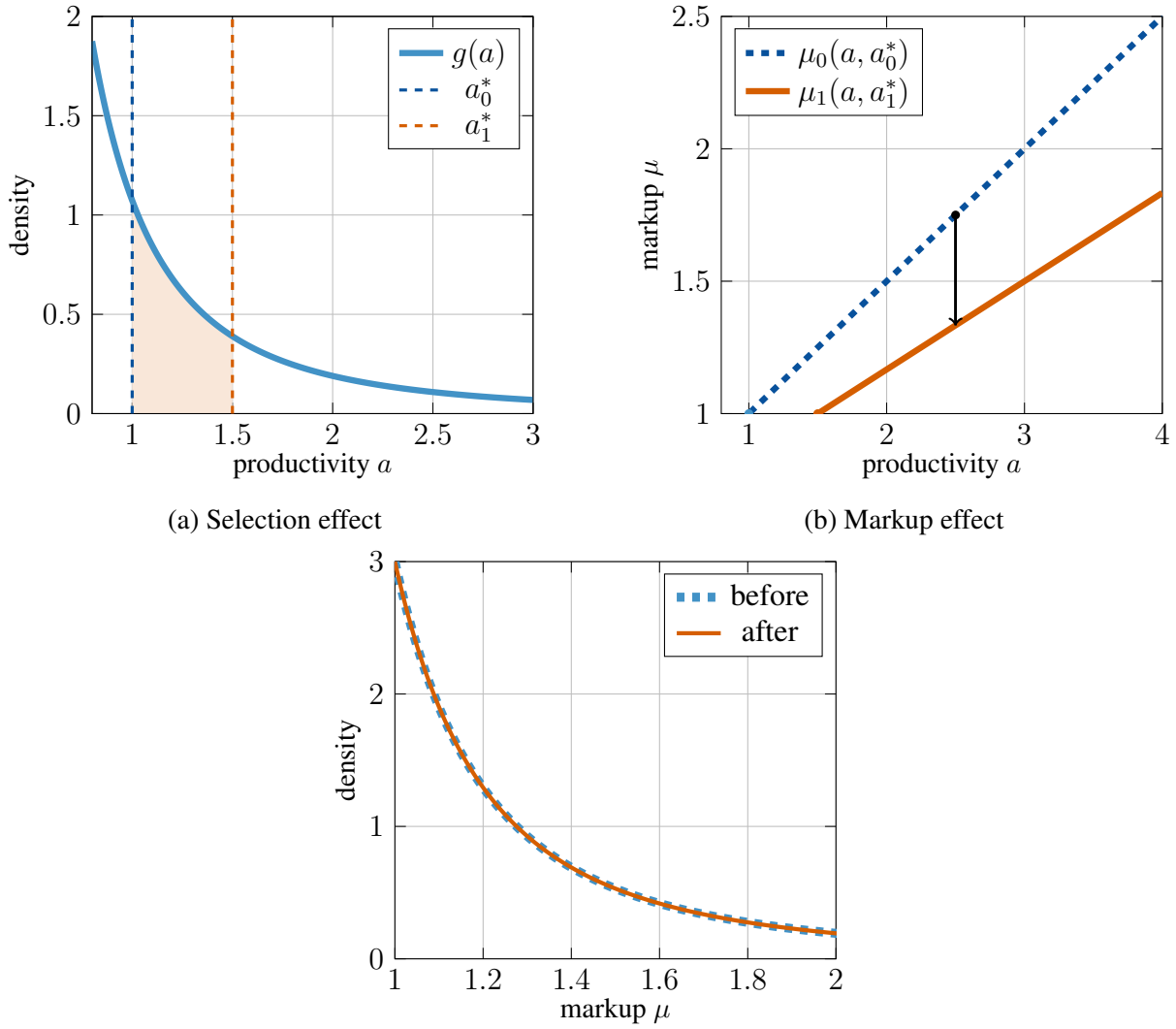
A 1% productivity shock induces a ρ_j-percent reduction in the sectoral price index P_j. For any finite $\kappa_j > 1$, pass-through is incomplete, $\rho_j < 1$. In the limiting case of $\kappa_j \to \infty$, $\rho_j \to 1$, corresponding to full pass-through.

The elasticity ρ_j measures how changes in production technology in sector j are transmitted into the sectoral price index. Growth pass-through is incomplete: a fraction of the productivity improvement is absorbed by entry and reallocation toward low-cost, high-markup firms rather than being fully transmitted into consumer prices. This mechanism relies on the variable elasticity of substitution across varieties and the resulting endogenous markups; it is absent in frameworks with CES and constant markups or in representative-firm model. The parameter κ_j governs the strength of the channel: when κ_j is small (a thick upper tail with many potential high-productivity firms), growth pass-through is weak, whereas for large κ_j (a thin tail with productivity bunched near the lower bound) it approaches one. This elasticity is the key to the structural change results later on: since ρ_j governs how productivity growth passes into the price index, it directly enters the relative price terms when determining expenditure shares across sectors.

A distinction arises between firm-level and sector-level pass-through. Firm-level marginal-cost pass-through is bounded above by one half, $\Phi_{ij} \in (0, 1/2)$, with the upper bound attained only by the marginal firm. Growth pass-through, by contrast, is bounded below by one half, $\rho_j \in (1/2, 1)$. The two elasticities describe different adjustment margins. Firm-level pass-through measures the response of an individual firm's price to marginal cost holding the sectoral choke price fixed, whereas growth pass-through captures the equilibrium response of the sectoral price index to a change in production technology. The latter additionally incorporates the endogenous adjustment of the productivity cut-off and firm selection induced by free entry.

An increase in the lower bound of the productivity distribution A_j shifts the entire distribution of potential technologies to the right and raises the equilibrium cut-off a_j^*. Figure 2 illustrates the resulting adjustment in firm selection and markups. As shown in Panel a, the higher cut-off causes low-productivity firms to exit, as represented by the shaded interval $[a_{j,0}^*, a_{j,1}^*)$. Moreover, for a given surviving incumbent with fixed productivity, the increase in a_j^* reduces its productivity relative to the cut-off and therefore lowers its markup. Panel b illustrates this intensive-margin response: the increase in the cut-off shifts the markup schedule downward, reducing the markup of a firm with fixed productivity $\bar{a}$.

At the same time, new firms entering from the new productivity distribution replace exiting low-productivity, low-markup producers. This selection effect shifts the composition of producers towards relatively more productive, higher-markup firms. As Panel c shows, the intensive-margin decline in incumbent markups and the selection effect exactly offset at the sector level. Conditional on producing, the distribution of relative productivity a/a_j^*, and therefore the distribution of firm-level markups, is invariant to a level shift in A_j. Consequently, the cost-weighted sectoral markup remains unchanged at $\mu_j^c = 1 + 1/\kappa_j$. Nevertheless, the endogenous adjustment in firm-level markups and firm composition prevents the productivity improvement from being transmitted fully into lower consumer prices, generating incomplete growth pass-through.



(a) Selection effect

(b) Markup effect

(c) Conditional markup distribution

Figure 2: Productivity Growth and Markup Invariance

Notes: Panel (a) plots the Pareto density of firm productivity together with the equilibrium cut-off before (a_0^*) and after (a_1^*) a productivity improvement; the shaded interval $[a_0^*, a_1^*)$ marks firms that were previously producing and are now below the cut-off. Panel (b) plots the markup schedule $\mu(a; a^*) = \frac{1}{2}(1 + a/a^*)$ before and after the shift; for a fixed incumbent $\bar{a}$ (surviving both regimes) the vertical segment shows the resulting decline in its markup. Panel (c) plots the density of the markup μ conditional on producing, implied by $\Pr(a/a^* \geq z \mid a \geq a^*) = z^{-\kappa_j}$ for $z \geq 1$; this density does not depend on A or a^*, so the curves coincide exactly, which is why the sectoral markup $\mu^c = 1 + 1/\kappa$ is unaffected by productivity improvements despite every surviving incumbent's markup falling.

2.4.3 Growth & Structural Change

Productivity growth implies falling prices. When $\eta < 1$, sectors are complements: a decline in P_j makes sector j cheaper, shifting relative expenditure to other sectors rather than pulling resources in. In a CES setting, this implies that sectors with faster productivity growth see their expenditure

share and labour input L_j fall relative to more "expensive" sectors.

With the nonhomothetic CES preferences, income effects also matter. Consider a one-time sector-specific growth shock that increases productivity from A_j to $A'_j > A_j$, holding A_k in all other sectors $k \neq j$ fixed. From the sectoral cut-off condition, the shock raises the cut-off a_j^* and reduces the sectoral price index P_j. The responses of expenditure share ω_j, employment L_j, and number of producing firms N_j depend on the sector's income elasticity ϵ_j, shape parameter κ_j, and can be summarised by the following proposition:

Proposition 2. *Consider an equilibrium described above (16)–(20) with non-homothetic CES preferences across sectors, $0 < \eta < 1$, and quadratic demand within sectors. Let A_j denote the lower bound of the productivity distribution in sector j, and let ω_j, L_j, and N_j be, respectively, the expenditure share, employment, and number of firms in sector j. Define the expenditure-weighted average income elasticity as $\bar{\epsilon} \equiv \sum_{m \in \mathcal{J}} \omega_m \epsilon_m$ and assume $\bar{\epsilon} > \eta$.*

A sector-specific technology improvement in sector j, modelled as an increase in A_j holding A_k fixed for all $k \neq j$, implies the following changes to expenditure shares and employment:

$$\begin{aligned} \frac{d \log \omega_j}{d \log A_j} = \frac{d \log L_j}{d \log A_j} &= \underbrace{-(1-\eta)(1-\omega_j)\rho_j}_{\textit{relative price effect}} + \underbrace{(\epsilon_j - \bar{\epsilon})\frac{d \log Q}{d \log A_j}}_{\textit{income effect}} \\ &= -(1-\eta)\rho_j \left[1 - \frac{\omega_j(\epsilon_j - \eta)}{\bar{\epsilon} - \eta}\right] \end{aligned} \tag{24}$$

$$\frac{d \log \omega_k}{d \log A_j} = \frac{d \log L_k}{d \log A_j} = \frac{d \log N_k}{d \log A_j} = (1-\eta)\rho_j \left[\frac{\omega_j(\epsilon_k - \eta)}{\bar{\epsilon} - \eta}\right], \qquad k \neq j \tag{25}$$

The number of producing firms in the own sector satisfies:

$$\frac{d \log N_j}{d \log A_j} = \frac{d \log \omega_j}{d \log A_j} + \rho_j = \rho_j \left[\eta + (1-\eta)\frac{\omega_j(\epsilon_j - \eta)}{\bar{\epsilon} - \eta}\right] \tag{26}$$

Proof. In Appendix A.3. ■

Equations (24) and (25) show that the response of sectoral expenditure shares to productivity growth is linear in growth pass-through ρ_j. For a given initial allocation and demand parameters, higher growth pass-through amplifies the magnitude of reallocation induced by productivity growth, while its direction depends on the sector considered. The change in expenditure shares can be decomposed into a relative price effect and an income effect. For the innovating sector j, the relative price effect is always negative when $\eta < 1$: a decline in the sectoral price index makes sec-

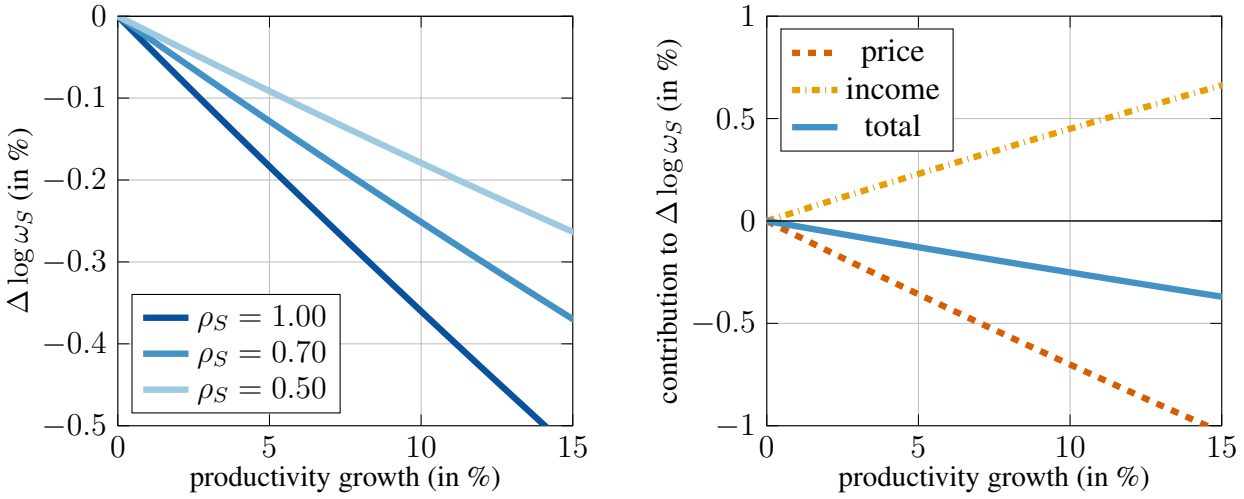
tor j relatively cheaper, and under gross complementarity this shifts expenditure towards the other sectors. The strength of this channel depends on growth pass-through ρ_j. The higher the growth pass-through, the larger the induced price movement and hence the stronger the price effect. The income effect depends on whether sector j is a relative necessity or a relative luxury. Since a productivity improvement raises aggregate real consumption, sectors with $\epsilon_k > \bar{\epsilon}$ gain expenditure share through the income channel, whereas sectors with $\epsilon_k < \bar{\epsilon}$ lose expenditure share. Hence, for an innovating necessity sector, both channels reduce its expenditure share, while for an innovating luxury sector the income effect works against the negative relative price effect and may offset it.

For the non-innovating sectors $k \neq j$, the relative price effect is always positive. Their prices are unchanged while the innovating sector becomes cheaper, so under complementarity expenditure is shifted towards the non-innovating sectors. The income effect is again governed by $\epsilon_k - \bar{\epsilon}$. Combining both channels, the net response is determined by $\epsilon_k - \eta$. The expenditure share of a non-innovating sector increases if $\epsilon_k > \eta$ and decreases if $\epsilon_k < \eta$. In the limiting case of $\eta \to 1$ and $\epsilon_j = 1 \quad \forall j \in J$, we are in the Cobb-Douglas setting and the expenditure shares are constant by construction.

Employment follows expenditure shares because $L_k = L\omega_k$. Hence, $\frac{d\log L_k}{d\log A_j} = \frac{d\log \omega_k}{d\log A_j}$ for all sectors. The response of the number of producing firms differs between the innovating and non-innovating sectors. In sectors $k \neq j$, productivity is unchanged and the number of producing firms therefore moves proportionally with sectoral employment, such that $\frac{d\log N_k}{d\log A_j} = \frac{d\log L_k}{d\log A_j}$. In the innovating sector, productivity growth additionally affects the extensive margin of production. The improvement in the productivity distribution raises the fraction of entrants that exceed the equilibrium productivity cut-off, implying $\frac{d\log N_j}{d\log A_j} = \frac{d\log \omega_j}{d\log A_j} + \rho_j$. Consequently, the number of producing firms in sector j may increase even when its expenditure share and employment decline.

Figure 3 illustrates how the degree of growth pass-through shapes expenditure reallocation following a productivity improvement in the Service sector. Panel 3a plots the total change in the Service expenditure share for three values of pass-through. Under complete pass-through, $\rho_S = 1$, productivity growth generates the largest decline in the Services expenditure share because the fall in the relative price of Services is strongest. Under gross complementarity, this shifts expenditure towards the other sectors. The reallocation shrinks with smaller growth pass-through. Panel 3b decomposes the case $\rho_S = 0.70$ into its two underlying channels. The relative-price effect is negative and increases in magnitude with the size of the productivity improvement, whereas the income effect is positive because Services have an income elasticity above the expenditure-weighted average, $\epsilon_S > \bar{\epsilon}$. The two forces therefore work in opposite directions and partially offset one another, leaving a smaller negative total response. Both the relative price and income channels

are proportional to growth pass-through ρ_S. Lower pass-through therefore reduces the magnitude of both channels and, holding the initial allocation fixed, scales down the overall expenditure-share response without changing its sign.



(a) Total response

(b) Decomposition $\rho_S = 0.70$

Figure 3: Services Productivity Growth and Expenditure Reallocation

Notes: The figure illustrates Proposition 2's decomposition of the change in expenditure share following a productivity improvement, at the example of Services. Panel (a) plots the total response $\Delta \log \omega_S$ implied by (24), for three values of the growth pass-through $\rho_S \in \{1.00, 0.70, 0.50\}$. Panel (b) decomposes the $\rho_S = 0.70$ case into its relative-price and income-effect components. The horizontal axis is the percentage increase in A_S. Parameter values are taken from the empirical section.

3 Data & Estimation

3.1 Data

World Input-Output Database. The World Input-Output Database (Timmer et al., 2015) provides information on final household consumption per industry, both in nominal values and previous-year prices.[9] I use this database to construct expenditure shares, relative prices and real consumption, to estimate the cross-sector parameters (ϵ, η). To obtain a long time series, I splice the Long-Run WIOD (1965–2000) with the WIOD 2016 Release (2000–2014). Because the two releases overlap in 2000, I retain the Long-Run WIOD observation in the overlapping year 2000. The industries are aggregated to sectors. A challenge in the aggregation process is the fact that the long-run WIOD and the 2016 release use a different industry classification system, namely ISIC Rev. 3.1 and ISIC

[9]The database is available from World Input-Output Database (2026).

Rev. 4, while also being partially pre-aggregated. I therefore rely on standard ISIC correspondence mappings (obtained from ILO) to align industries across revisions wherever feasible. I split up sector "[J] Information and Communication" (ISIC Rev. 4) and allocate parts of it to "Manufacturing", "Transport & Communication", and "Other Services" to maintain comparability with Rev. 3.1 from the Long-Run WIOD. Table B.1 summarises the aggregation procedure from ISIC Rev. 3.1 and Rev. 4. This leaves me with 8 consistently defined sectors on the lower level of aggregation. Then, the 8 sectors are further aggregated into 3 broad sectors Agriculture, Industry, and Services. The population data to calculate real consumption per capita comes from the 11th issue of the Penn World Tables (Feenstra et al., 2015).

Compustat. For firm-level data, I use Compustat.[10] Specifically, I employ the Compustat North America Fundamentals Annual file, which contains all US-incorporated publicly listed firms. Compustat provides financial information using balance sheet and income statements. This entails, among other things, information on sales, cost of goods sold (COGS), capital stock, employment, and industry activity classification.[11] My sample covers all US-domestic firms from 1980-2014 with a unique firm identifier. To select US firms only, I filter on ISO country code and ISO currency code. I drop all observations without industry assignment (missing NAICS code), as well as missing or negative entries for sales or COGS. Following Traina (2018), I remove big merger and acquisition (M&A) episodes. That is, I drop observations if M&A surpasses 5% of total assets. Following De Loecker et al. (2020), I exclude observations in the top and bottom 1% of COGS-to-sales distribution each year. I assign each observation to a 2-digit NAICS industry, which leaves 20 distinct industries in the final sample.[12] See Table 1 for summary statistics of the sample. Table C.1 in the appendix shows the mapping from the 2-digit NAICS industries into sectors.

[10] The Compustat data is available from Wharton Research Data Services (2026).

[11] Compustat has two important limitations for markup estimation. First, it covers publicly listed firms rather than the full universe of firms. As emphasized by De Loecker et al. (2020), listed firms are not representative of the overall firm distribution: they are larger, older, more capital- and skill-intensive, and the industry mix differs from that of the private sector. Traina likewise stresses that public-firm samples can overstate aggregate market power because public firms are larger and their representativeness changes over time with listings, delistings, and mergers (Traina, 2018). Second, Compustat provides revenues and expenditures, but not firm-level prices and quantities. As a result, markup estimation must rely on the production approach using revenue data rather than direct price–quantity information. Recent work highlights that this can be problematic because revenue-based estimates may conflate prices, quantities, quality, and demand shifters, and different flexible inputs need not recover the same markup (Raval, 2023; Bond et al., 2021; De Ridder et al., 2026). By contrast, Burstein et al. (2025) show that using firm-level quantity information instead of revenues can circumvent some of these biases.

[12] I merge NAICS industry 49 into 48 and 61 into 62 due to limited sample sizes.

Table 1: Summary Statistics Compustat (1980-2014)

Variable	Mean	Median	Observations
Sales	1,291.5	81.6	213,703
COGS	881.9	47.4	213,703
SGA	218.9	18.0	171,613
Employment	5.9	0.5	192,455

Notes: The table reports summary statistics for the cleaned Compustat firm-year sample used in the markup estimation. Sales, COGS, and SG&A are measured in millions of U.S. dollars. Employment is measured in thousands of employees.

3.2 Cross-Sector Parameter Estimation

To estimate the cross-sector parameters $(\boldsymbol{\epsilon}, \eta)$, I make use of (6) and the WIOD. Expenditure shares per sector are constructed using final household consumption. The expenditure exists both in nominal terms as well as in previous-year prices. Hence, I can infer the year-on-year price change factor for each sector as $g^P_{j,t} = E^{\text{nom}}_{j,t} / E^{\text{pyp}}_{j,t}$, where $E^{\text{nom}}_{j,t}$ denotes nominal final household expenditure in sector j in year t, and $E^{\text{pyp}}_{j,t}$ denotes expenditure on the same quantities valued at previous-year-prices.

WIOD reports final consumption by country of origin, which allows me to construct both domestic and trade-inclusive time series of relative prices and expenditure shares. The trade-inclusive expenditure share is total household spending on a sector regardless of where it was produced, thus giving a more accurate reflection of household consumption patterns. The implicit price deflators can be constructed accordingly. I take the trade-inclusive specification as baseline throughout, and report the domestic-only results as a robustness check.

Appendix B.2 and B.3 show the expenditure shares and relative prices across 3-sector and 8-sector aggregates. In the 3-sector aggregation, the continuation across WIOD releases is very close. In the 8-sector aggregation, there are discontinuities around the splice in some sectors. Therefore, I discard the 2000-2001 first-difference observations in the 8-sector aggregation to avoid contaminating the estimates with these discontinuities.

Following Comin et al. (2021), the estimation procedure requires a base sector b, with the normalisation of $\epsilon_b = 1$, and all other income elasticities are interpreted as relative to the base sector's income elasticity. I use Industry as base sector. Because WIOD does not provide price indices, only price changes, I rewrite (6) in changes rather than levels:

$$\Delta \log \left(\frac{\omega_{j,t}}{\omega_{b,t}} \right) = (1 - \eta) \, \Delta \log \left(\frac{P_{j,t}}{P_{b,t}} \right) + (\epsilon_j - 1) \, \Delta \log Q_t \tag{27}$$

Using $g^P_{j,t}$, relative price changes can be expressed as

$$\begin{aligned}\Delta \log\left(\frac{P_{j,t}}{P_{b,t}}\right) &= (\log P_{j,t} - \log P_{j,t-1}) - (\log P_{b,t} - \log P_{b,t-1}) \\ &= \log\left(\frac{g^P_{j,t}}{g^P_{b,t}}\right) = \log\left(\frac{E^{\text{nom}}_{j,t}/E^{\text{pyp}}_{j,t}}{E^{\text{nom}}_{b,t}/E^{\text{pyp}}_{b,t}}\right),\end{aligned}$$

and real consumption is defined as aggregate nominal household consumption deflated by the price level, $Q_t \equiv E_t/P_t$ with $E_t \equiv \sum_{m\in\mathcal{J}} E^{\text{nom}}_{m,t}$. Since the aggregate price level is unobserved, I substitute it with indirect measurement using (5), implying

$$\Delta \log Q_t = \Delta \log E_t - \Delta \log P_{b,t} + \frac{1}{1-\eta}\Delta \log \omega_{b,t} = \Delta \log E_t - \log g^P_{b,t} + \frac{1}{1-\eta}\Delta \log \omega_{b,t}$$

Finally, differencing log relative expenditure shares removes time-invariant sector-specific components, so sector fixed effects cancel mechanically in the transformed equation.

I estimate the differenced cross-sector demand system by non-linear GMM. Appendix B.4 provides details about the procedure. The estimates are reported in Table 2. The income elasticities are reported as $\epsilon_j - 1$, i.e. relative to the base sector. The results are consistent with the standard structural-change pattern. Agriculture has an income elasticity below that of Industry, with $\epsilon_A - 1 = -0.36$ in the domestic specification and $\epsilon_A - 1 = -0.13$ in the trade-inclusive specification, implying that Agriculture behaves as a relative necessity good. In Services, the point estimates are 1.39 and 0.64, respectively, making Services a relative luxury good. The estimated substitution parameter η lies between 0.33 and 0.53 across specifications, indicating imperfect substitutability across broad sectors, consistent with my assumption in the model. Overall, the estimates support the view that non-homothetic demand is an important force behind structural change, with Agriculture as a necessity sector and Services as a luxury sector.

3.3 Measuring Markups

I construct firm-level markups following the production approach of De Loecker et al. (2020). Under cost minimization and competitive input markets, a firm's markup can be recovered from the ratio of the output elasticity of a flexible input to that input's revenue share. Let $R_{it} = P_{it}Q_{it}$ denote firm sales and let C_{it} denote expenditure on the variable input bundle. I use cost of goods sold (COGS) as the measure of variable input expenditure. For firm i in industry n and year t, the

Table 2: Cross Sector Parameter Estimates: 3-Sector Aggregation

	(1)	(2)
η	0.33	0.53
	[0.27,0.38]	[0.52,0.55]
Agriculture	-0.36	-0.13
	[-0.46,-0.27]	[-0.14,-0.11]
Services	1.39	0.64
	[1.01,1.78]	[0.58,0.70]
Sector FE	N	N
Trade Controls	N	Y
Observations	98	98

Notes: The table reports non-linear GMM estimates of the differenced relative expenditure-share equation. Column (1) uses domestic expenditure shares and domestic implicit deflators only. Column (2) uses trade-inclusive expenditure shares and implicit deflators, i.e. imports are included in household expenditure and price measurement. The base sector is industry, so $\epsilon_b = 1$ by normalisation; hence the reported sector coefficients are estimates of $\epsilon_j - 1$ relative to the base sector. Brackets report 95% confidence intervals. Observations are sector-year cells. Because the estimating equation is written in first differences, time-invariant sector fixed effects difference out mechanically; therefore no sector fixed effects are included.

markup is then given by

$$\mu_{it} = \theta^V_{nt} \frac{R_{it}}{C_{it}} \tag{28}$$

where θ^V_{nt} is the output elasticity of the variable input bundle (see Appendix C.2 for a derivation of this markup formula).[13] Rather than re-estimating these elasticities, I take the industry-year output elasticities directly from De Loecker et al. (2020) and merge them to the Compustat firm-year data by year and two-digit NAICS Industry.[14] Firm-level variation in markups therefore comes from differences in firms' inverse variable input shares, R_{it}/C_{it}, while the output elasticity is common within Industry-year cells.

For sector-level markups, I aggregate firm-level markups using variable-cost weights. Let $\mathcal{F}_j$

[13]The production approach to markup estimation is subject to well-known critiques. Raval (2023) shows that markup estimates differ substantially depending on the input factor used for estimation — COGS, as used here following De Loecker et al. (2020), versus alternatives such as labour or materials. Traina (2018) shows that correctly accounting for indirect costs such as marketing and management substantially reduces the estimated increase in average markups. See Basu (2019) for a general discussion.

[14]De Loecker et al. (2020) estimate the output elasticity of the variable input using industry-specific production functions. In their benchmark specification, log output is modelled as a Cobb-Douglas function of a variable input bundle and capital, $q_{it} = \beta_v v_{it} + \beta_k k_{it} + \omega_{it} + \varepsilon_{it}$, where output is measured using deflated firm sales and the variable input bundle is proxied by cost of goods sold. To address the simultaneity between input choices and unobserved productivity, they use a control-function approach. In the first stage, they purge measurement error and unanticipated shocks to sales by estimating output as a flexible function of variable inputs, capital, and year effects. Productivity is then assumed to follow an AR(1) process, and the output elasticity of the variable input is identified from moments based on lagged variable input choices. I use their estimated industry-year elasticities directly rather than re-estimating the production functions.

denote the set of Compustat firms assigned to broad sector j. For sector j, the cost-weighted markup is

$$\mu_{j,t} = \frac{\sum_{i \in \mathcal{F}_j} C_{it} \mu_{it}}{\sum_{i \in \mathcal{F}_j} C_{it}} \tag{29}$$

I construct this measure both for two-digit NAICS industries and the sector aggregates. The aggregate markup is obtained analogously by applying the same cost-weighted average across all firms in the cleaned Compustat sample. Appendix C.3 plots the evolution of the aggregate markup from 1980 to 2014 in the US economy. The aggregate markup increased from 1.13 in 1980 to 1.24 in 2014. Appendix C.4 plots the average markups in the 3 sectors.

Table 3 reports initial and final values of aggregate and sector markups from 1980 to 2014, as well as the percentage change. Across sectors, the largest percentage increase is observed in Agriculture, where the average markup rises from 1.11 to 1.59. Markups in Industry increase by 11.8% from 1.13 to 1.26, whereas Services markups grow by 6.8%, from 1.15 to 1.23.

Table 3: Average Markups by Sector

	Average markup		
Sector	**1980**	**2014**	Δ
Aggregate	1.13	1.24	9.4%
Agriculture	1.11	1.59	43.5%
Industry	1.13	1.26	11.8%
Services	1.15	1.23	6.8%

Notes: Average markups are computed as COGS-weighted averages of firm-level markups, using Compustat data and output elasticities from De Loecker et al. (2020). The table reports initial and final values of sectoral markups, as well as the percentage change. The final value of Agriculture's markup is replaced by the average of its last three values to reduce endpoint volatility.

Using (22), I can decompose the change in aggregate markup into changes in sector sizes and changes in sector markups. Appendix C.5 shows that, despite the large increase in agricultural markups, Agriculture is essentially irrelevant for the aggregate markup because its share in aggregate COGS is negligible. Instead, the rise in the aggregate markup is driven primarily by Services. Services contribute through both an increase in the markup and, more importantly, the reallocation of economic activity toward Services, as their COGS share increases to over 50%. The markup increase in Industry is overturned by its declining size. This pattern is in line with Marto (2024), who emphasizes that the rise in aggregate markups is partly driven by structural change itself.

4 Calibration

Section 3 estimated the cross-sector demand parameters and constructed firm- and sector-level markups. I now use these empirical objects to take the structural model to the data. First, I recover the sectoral productivity distributions and productivity paths from observed markups and prices, and use the free-entry condition to infer the underlying technology paths. The recovered productivity distributions also determine sectoral growth pass-through. I then calibrate the model to the 1980 expenditure shares. Given the estimated demand parameters and observed sectoral prices, the model closely tracks the untargeted reallocation away from Agriculture and Industry and towards Services.

4.1 Calibration Setup

4.1.1 Firm Productivity Distribution

The shape parameter $\kappa_{j,t}$ of the sectoral firm productivity distribution is pinned down by observed sectoral markups. By (21), the sectoral markup satisfies

$$\mu^c_{j,t} = 1 + \frac{1}{\kappa_{j,t}} \qquad \Leftrightarrow \qquad \kappa_{j,t} = \frac{1}{\mu^c_{j,t} - 1} \tag{30}$$

Thus, changes in sectoral markups are interpreted as changes in the shape of the underlying productivity distribution. The model requires $\kappa_{j,t} > 1$, corresponding to $1 < \mu^c_{j,t} < 2$. Since annual Compustat markups contain substantial short-run variation, I do not use the $\kappa_{j,t}$ series directly. Instead, I construct a smoothed path that exactly matches the values in 1980 and 2014 and assumes that $\kappa_{j,t} - 1$ evolves exponentially between the two endpoints,

$$\kappa_{j,t} = 1 + (\kappa_{j,1980} - 1) \exp\left[-\zeta_j (t - 1980)\right], \tag{31}$$

where ζ_j is the decay rate, chosen such that the path matches the markup-implied value of $\kappa_{j,2014}$. This preserves the observed long-run change in sectoral markups while abstracting from short-run fluctuations.[15]

The calibrated firm productivity distributions also determine sectoral growth pass-through. By equation (23), growth pass-through is given by $\rho_{j,t} = \kappa_{j,t}/(1 + \kappa_{j,t})$. Table 4 reports the ini-

[15] Results are robust to alternative paths of $\kappa_{j,t}$. Using linear interpolation between endpoints or the annual markup-implied series instead of the exponential path leaves the effect of full pass-through on structural change unaltered. Naturally, using the annual series reproduces short-run movements in the aggregate markup, as these fluctuations are directly fed into the calibration.

tial and final values of $\kappa_{j,t}$ together with the corresponding model-implied growth pass-through. Pass-through declines in all three sectors, from 90% to 63% in Agriculture, from 89% to 79% in Industry, and from 87% to 82% in Services. The decline is by far the most pronounced in Agriculture, consistent with the largest fall in $\kappa_{j,t}$ and largest increase in its sectoral markup, as documented above.

Table 4: Growth Pass-Through

Sector	κ_{1980}	κ_{2014}	ρ_{1980}	ρ_{2014}
Agriculture	9.2	1.7	90%	63%
Industry	7.9	3.9	89%	79%
Services	6.7	4.4	87%	82%

Notes: Cost-weighted markups imply $\kappa_{j,t} = 1/(\mu^c_{j,t} - 1)$. The model growth pass-through elasticity is $\rho_{j,t} = \kappa_{j,t}/(1 + \kappa_{j,t})$, holding κ fixed in the comparative static. The baseline uses the smoothed kappa series.

4.1.2 Productivity Cut-off

Given the calibrated firm productivity distributions, the evolution of the sectoral productivity cut-off can be recovered from WIOD sectoral price changes. By (20), sectoral price changes can be decomposed into:

$$\Delta \log P_{j,t} = \Delta \log \left(\frac{\kappa_{j,t} + 1}{\kappa_{j,t} + 2} \right) - \Delta \log a^*_{j,t}$$

Rearranging gives the implied change in the productivity cut-off:

$$\Delta \log a^*_{j,t} = \Delta \log \left(\frac{\kappa_{j,t} + 1}{\kappa_{j,t} + 2} \right) - \Delta \log P^{data}_{j,t} + \Delta \log w_t \tag{32}$$

I adjust prices by aggregate wage growth. This is necessary because the model holds the wage fixed by normalisation, whereas observed price changes in the data reflect both changes in productivity and nominal labour costs. I use the BLS non-farm business hourly compensation index from FRED (*COMPNFB*) to measure $\Delta \log w_t$. This adjustment was not necessary previously, since wages cancel in relative prices, which were used to estimate cross-sector parameters.

Column 1 of Table 5 reports the average annual log change of the productivity cut-off $a^*_{j,t}$ over the sample period from 1980 to 2014, using the trade-inclusive price series of the baseline specification. The cut-off increases most rapidly in Agriculture (2.7%) and Industry (2.4%), while the

corresponding increase in Services is only 0.6%. These cut-off productivity paths are subsequently used to recover the underlying technology paths.

To get a better sense of the average productivity of producing firms, Table 5 also reports a distributional effect depending on $\kappa_{j,t}$ (Column 2) and the change in the conditional-mean productivity $E[a_{j,t} \mid a_{j,t} \geq a^*_{j,t}]$ (Column 3). The conditional mean of a Pareto distribution is given by:

$$E[a_{j,t} \mid a_{j,t} \geq a^*_{j,t}] = a^*_{j,t} \cdot \frac{\kappa_{j,t}}{\kappa_{j,t} - 1}$$

Hence, growth in mean productivity among producing firms can be decomposed into a cut-off effect and a distributional effect. Average productivity growth rates are 5.0% in Agriculture, 2.9% in Industry, and 0.9% in Services (see Appendix D.1 for the decomposition using domestic values). The distributional effect is largest in Agriculture and smallest in Services, consistent with the sectoral differences in markup growth and the corresponding changes in $\kappa_{j,t}$. Low $\kappa_{j,t}$ implies a fat upper tail, hence there are more high-productivity firms among producing firms, raising average productivity.

Table 5: Productivity Growth Decomposition

Sector	**Cut-off a^***	**Distribution**	**Total**
Agriculture	2.7%	2.3%	5.0%
Industry	2.4%	0.5%	2.9%
Services	0.6%	0.3%	0.9%

Notes: Values are average annual log changes. Conditional-mean productivity among producing firms, $E[a_{j,t} \mid a_{j,t} \geq a^*_{j,t}] = a^*_{j,t} \cdot \kappa_{j,t}/(\kappa_{j,t} - 1)$, is decomposed into $\Delta \log E[a_{j,t} \mid a_{j,t} \geq a^*_{j,t}] = \Delta \log a^*_{j,t} + \Delta \log[\kappa_{j,t}/(\kappa_{j,t} - 1)]$. The cut-off effect captures the shift in the productivity threshold of producing firms; the distribution effect captures the change in the shape of the productivity distribution above that threshold. Since this only depends on $\kappa_{j,t}$, it is identical across the domestic and trade-inclusive specifications. The baseline uses the smoothed series for κ.

The distributional term in this decomposition should be distinguished from the $\kappa_{j,t}$-dependent term in equation (32). The former, $\Delta \log[\kappa_{j,t}/(\kappa_{j,t} - 1)]$, measures how changes in the shape of the productivity distribution affect mean productivity among producing firms. The latter, $\Delta \log[(\kappa_{j,t} + 1)/(\kappa_{j,t} + 2)]$, captures the corresponding contribution of $\kappa_{j,t}$ to price growth. Appendix D.2 reports the corresponding price decomposition, while Appendix D.3 plots the evolution of the productivity cut-off $a^*_{j,t}$ against the inverse price index per sector. A decline in $\kappa_{j,t}$ therefore raises mean productivity relative to the cut-off by increasing the weight of high-productivity firms, while also lowering the sectoral price index for a given productivity cut-off.

4.1.3 Technology Frontier

The model's free-entry condition relates the productivity cut-off $a^*_{j,t}$ to the underlying technology frontier $A_{j,t}$ through:

$$\log a^*_{j,t} = \frac{\kappa_{j,t}}{1+\kappa_{j,t}} \log A_{j,t} - \frac{K + \log[(\kappa_{j,t}+1)(\kappa_{j,t}+2)]}{1+\kappa_{j,t}} \tag{33}$$

where $K \equiv \log(2\gamma F/(L\alpha))$ collects the entry cost, market size, and lower-tier demand parameters. Given the calibrated paths of $\kappa_{j,t}$ and the recovered productivity cut-off paths $a^*_{j,t}$, (33) can be inverted to recover $A_{j,t}$. Since the data identify changes in $a^*_{j,t}$ but not levels, an additional normalization is required. In the baseline, I normalize $a^*_{j,1980} = 1$ for all sectors. K is not separately identified. I set $K = 0.0$ as my baseline and report the sensitivity of the results to K in Appendix D.4.

4.1.4 Sectoral Demand Weights

Since the relative expenditure share equation (6) is identified in changes, I normalize sectoral relative prices and the aggregate consumption index Q_t to one in the base year. Under this normalization, the model implies $\omega_{j,1980} = \beta_j$, so I calibrate β_j to match the observed 1980 expenditure share of each sector. The resulting preference weights are reported alongside the other calibration parameters in Table 6.

4.2 Baseline Results

Table 6 summarizes the parameters used in the baseline calibration. The cross-sector demand parameters η and ϵ_j are estimated using WIOD data in Section 3, while the preference weights β_j are calibrated to the observed expenditure shares in 1980. The decay rates ζ_j govern the smoothed paths of the Pareto shape parameters and are chosen to match the values of $\kappa_{j,t}$ implied by sectoral markups in 1980 and 2014.

The baseline model is disciplined by observed sectoral prices and markups. Sectoral markups determine the productivity distributions, while observed price changes are used to recover the productivity cut-offs and, through the free-entry condition, the underlying technology paths. Since sectoral price changes are reproduced by construction, the remaining discrepancies between observed and model-implied expenditure shares reflect the fit of the estimated upper-tier demand system.

Table 6: Calibration Parameters

Parameter	η	ϵ_A	ϵ_I	ϵ_S	β_A	β_I	β_S	ζ_A	ζ_I	ζ_S
Value	0.53	0.87	1.00	1.64	0.01	0.28	0.71	0.073	0.026	0.015

Notes: The table reports the trade-inclusive upper-tier demand parameters and the sector-specific decay rate of the smoothed kappa series. η is the elasticity of substitution across sectors; the sectoral income elasticities ϵ_j are estimated using trade-inclusive expenditure shares and implicit deflators, including imports, with the Industry income elasticity normalized to one. The preference weights β_j are calibrated to the observed 1980 expenditure shares, as relative sectoral prices and aggregate consumption are normalized to one in that year. ζ_j governs the exponential decay of $\kappa_{j,t} - 1$, with the smoothed paths matching the markup-implied values in 1980 and 2014.

The results closely track the long-run reallocation of expenditure away from Agriculture and Industry and toward Services. Table 7 confirms the close fit: the correlations between observed and model-implied expenditure shares are 0.97 in Agriculture, 0.98 in Industry, and 0.98 in Services, while the RMSE amounts to 6.0%, 3.1%, and 0.9% of the respective average sectoral expenditure shares. Appendix D.6 provides a visual comparison of observed and calibrated expenditure shares.

Table 7: Baseline Fit of Expenditure Shares, 1980–2014

Sector	**RMSE**	**% of share**	**Correlation**
Agriculture	0.0005	6.0%	0.97
Industry	0.0069	3.1%	0.98
Services	0.0071	0.9%	0.98

Notes: Baseline calibration of the model. Empirical paths for the Pareto shape parameter $\kappa_{j,t}$ and effective productivity $a^*_{j,t}$ are used to recover the underlying technology shifter $A_{j,t}$ through the free-entry condition. Expenditure shares are obtained from the upper-tier demand system. Since baseline sectoral prices are taken from the data, remaining share discrepancies reflect demand calibration rather than errors in matching prices.

Figure 4 compares the aggregate markup in the data and the calibrated baseline. The calibrated markup does not reproduce the short-run fluctuations in the observed series due to the smoothing of $\kappa_{j,t}$. However, it captures the secular trend over the sample period and closely matches the observed level at the end of the sample. Since the aggregate markup also depends on sectoral COGS shares, its close fit provides additional confirmation that the calibrated model captures the observed structural reallocation.

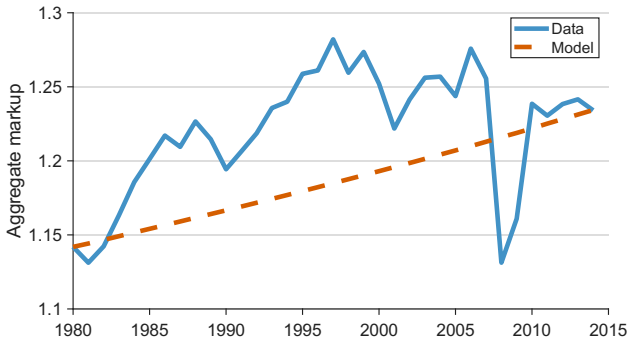


Figure 4: Calibrated Aggregate Markup

Notes: The figure compares the data-implied and calibrated aggregate markups. Both are constructed using the model's aggregation formula, ($\mathcal{M}_t = [\sum_j \omega_{j,t}/\mu^c_{j,t}]^{-1}$). The data-implied series combines observed WIOD expenditure shares with Compustat sectoral markups. The calibrated series uses model-implied expenditure shares and sectoral markups obtained from the smoothed Pareto shape parameters.

5 Counterfactual Growth Pass-Through

5.1 Full Growth Pass-Through Counterfactual

I use the calibrated model to quantify the effect of limited growth pass-through on structural change. I consider a counterfactual with full growth pass-through, as commonly assumed in the structural change literature with representative firms and constant markups. In my model, this corresponds to the limit $\kappa_{j,t} \to \infty$, in which sectoral markups converge to one and the sectoral price index becomes $P_{j,t} = 1/A_{j,t}$. Thus, changes in the technology frontier are passed through one-for-one into sectoral prices. By construction, sectoral markups are equal to one throughout the full pass-through counterfactual, implying an aggregate markup of one. I construct the counterfactual by holding fixed the recovered paths of $A_{j,t}$, imposing complete pass-through, and re-solving the demand system for the resulting counterfactual prices, aggregate consumption, and expenditure shares. Comparing these counterfactual shares with the baseline isolates the contribution of limited growth pass-through to sectoral reallocation.

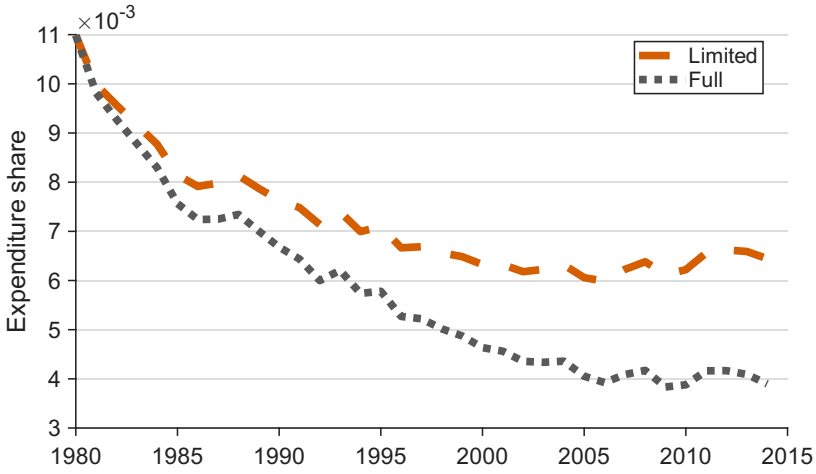
Figure 5 compares the calibrated limited pass-through baseline and the full pass-through counterfactual. The full pass-through counterfactual shows that limited pass-through substantially dampens structural change. By 2014, imposing full pass-through lowers the expenditure and employment share of Industry by 2.67 percentage points and raises the corresponding share of Services by 2.92 percentage points relative to the baseline. The additional 2.92pp expansion of Services under full pass-through corresponds to *31.4% of the entire 1980–2014 increase* in the model-implied Services expenditure share. Put differently, structural change toward Services would have been 31.4% larger under full pass-through. Given total US employment of about 146

million workers in 2014, this would correspond to approximately 4.3 million additional workers in Services. The effect on Agriculture is small in absolute terms, at -0.25pp, but sizeable relative to its baseline structural change: under full pass-through, the decline in the Agriculture share would have been 56.1% larger.

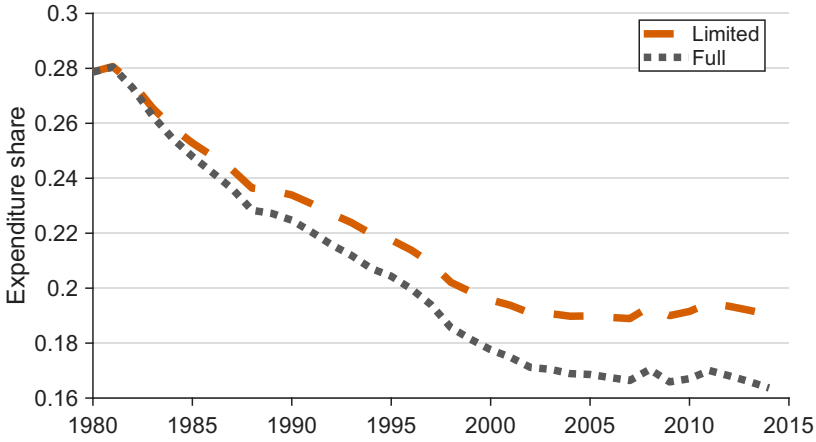
5.2 Price & Income Effects Decomposition

Table 8 decomposes the change in relative expenditure shares into price and income effects using (6). In the baseline, income effects account for 27% of the total change in Agriculture relative to Industry and for 41% in Services relative to Industry. Price effects therefore account for the bulk of structural change in both cases, although the two channels are more balanced for Services. This differs from Comin et al. (2021), who find that income effects dominate within-country structural change in their cross-country sample. Their income channel accounts for more than 73% of sectoral reallocation. By contrast, Boppart (2014) finds roughly equally large income and price effects for the US. The price effect column can equivalently be interpreted as the structural change response in a homothetic specification with $\epsilon_j = 1$ for all sectors, holding the estimated elasticity of substitution and sectoral price paths fixed.

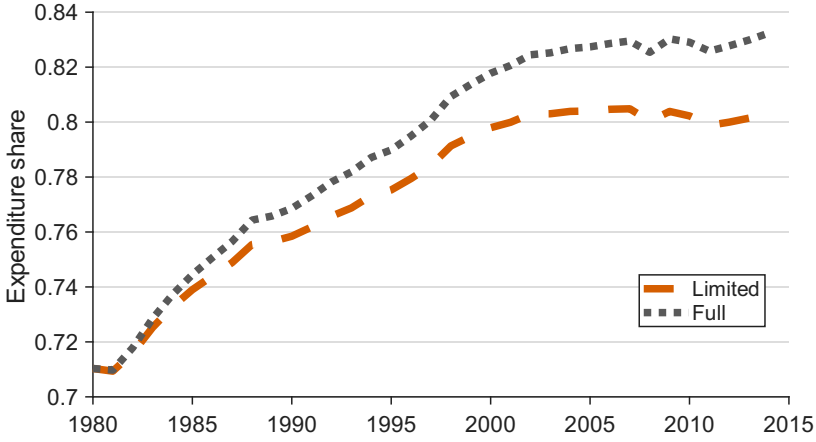
Imposing full pass-through strengthens both channels, but the additional structural reallocation operates primarily through relative prices. For Agriculture, the price effect increases sharply in magnitude to -0.45, while the income effect increases only modestly to -0.05, reducing the income contribution to 11%. For Services, the price effect rises to 0.41 and the income effect to 0.28. The stronger income effect reflects the increase in aggregate real consumption under full pass-through. As shown in Appendix D.5, the baseline and counterfactual consumption paths diverge over time, with real consumption reaching about 1.55 under full pass-through compared with 1.39 under limited pass-through by 2014. Nevertheless, the increase in the price effect is quantitatively larger than the increase in the income effect. Hence, limited growth pass-through slows structural change primarily by dampening relative price changes, while the associated reduction in real consumption further weakens the income channel.



(a) Agriculture



(b) Industry



(c) Services

Figure 5: Expenditure Shares: Limited versus Full Growth Pass-Through

Notes: Each panel compares the model-implied share under the limited pass-through baseline and the full pass-through counterfactual.

Table 8: Decomposition: Price vs. Income Effects

	Price effect	**Income effect**	**Income share**
Baseline			
Agriculture vs. Industry	-0.11	-0.04	27%
Services vs. Industry	0.30	0.21	41%
Counterfactual			
Agriculture vs. Industry	-0.45	-0.05	11%
Services vs. Industry	0.41	0.28	40%

Notes: Decomposition of $\Delta \log(\omega_j/\omega_{Industry})$, 1980–2014, into $(1-\eta)\Delta \log(P_j/P_{Industry})$ and $(\epsilon_j - 1)\Delta \log Q$ following (6).

5.3 Sectoral Decomposition

To identify which sectors drive the aggregate effect of limited growth pass-through, I impose full pass-through sector by sector and for all possible combinations. Table 9 Panel A reports the resulting decompositions. The results show that the aggregate effect is driven almost entirely by Industry. Imposing full pass-through in Industry alone lowers its expenditure share by 3.21 percentage points and raises the Services share by 3.20 percentage points by 2014, close to the 2.92 percentage-point increase obtained when full pass-through is imposed in all sectors. In contrast, imposing full pass-through only in Services lowers its share by 0.63 percentage points. This contrast reflects the relative price mechanism under gross complementarity: stronger pass-through in an innovating sector translates productivity growth into a larger decline in its relative price and thereby reduces its own expenditure share. The Shapley decomposition in Panel B confirms this result, attributing 3.25 percentage points of the overall Services-share effect to Industry, compared with 0.25 percentage points to Agriculture and -0.58 percentage points to Services. Hence, market power slows the transition toward Services primarily by preventing productivity growth in Industry from translating into lower industrial prices, rather than by limiting price declines within Services itself.

Table 9: Sectoral Decomposition of Full Pass-Through Counterfactual

	2014 share gap (in pp)		
Counterfactual	**Agriculture**	**Industry**	**Services**
Panel A: Combinations			
None (baseline)	0.00	0.00	0.00
Agriculture	-0.27	0.02	0.24
Industry	0.01	-3.21	3.20
Services	0.02	0.61	-0.63
Agriculture + Industry	-0.26	-3.19	3.45
Agriculture + Services	-0.26	0.64	-0.38
Industry + Services	0.02	-2.69	2.66
All sectors	-0.25	-2.67	2.92
Panel B: Shapley Attribution			
Agriculture	-0.27	0.02	0.25
Industry	0.01	-3.25	3.25
Services	0.01	0.57	-0.58
Sum	-0.25	-2.67	2.92

Notes: Panel A reports the 2014 expenditure-share gap relative to the limited-pass-through baseline when full growth pass-through is imposed in the indicated sectors. The recovered underlying technology paths $A_{j,t}$ are held fixed, untreated sectors retain their baseline price paths, and aggregate real consumption and expenditure shares are re-solved in each counterfactual. Panel B reports the Shapley decomposition of the all-sector full-pass-through effect, averaging each sector's incremental contribution over all possible orders in which sectors are switched to full pass-through. The Shapley contributions sum to the all-sector effect.

6 Extensions

6.1 Disaggregating Sectors

The main analysis studies structural change across three broad sectors. To assess heterogeneity within these aggregates, I disaggregate the economy into eight sectors. Tables E.1 and E.2 report the corresponding estimates of the cross-sector demand parameters, productivity growth, and growth pass-through. On the demand side, the high income elasticity of the Service sector is driven primarily by Finance and Transport & Communication, with relative income elasticities near or above 1. Business Services and Other Services are also more income elastic than Manufacturing, although by smaller margins. The relative income elasticity of Other Industry (including Mining, Utilities, and Construction), smaller than Manufacturing. The point estimates for Agriculture are virtually identical to before.

The supply side exhibits equally pronounced heterogeneity. Trade Services combine relatively

strong effective-productivity growth of 1.7% per year with essentially unchanged growth pass-through of about 90%. Transport & Communication also experiences strong productivity growth, at 2.0%, but pass-through declines from 92% to 85%. Finance provides a particularly interesting contrast: despite being among the most income-elastic sectors, productivity growth is only 0.5%, while pass-through falls substantially from 88% to 70%. Business Services display similarly modest productivity growth but a much smaller decline in pass-through. At the other end, Other Services experience negative measured productivity growth and persistently low pass-through. This residual category contains many less market-oriented activities, including public administration, education, health, and other social and personal Services. The results do not suggest a simple marketable–non-market Services divide. Instead, substantial heterogeneity exists even among marketable Services: Finance, Transport, Trade, and Business Services differ markedly in both their income elasticities and the extent to which productivity growth is transmitted into prices.

These patterns are consistent with the importance of heterogeneity within Services emphasized by Duernecker et al. (2024). They distinguish between progressive and stagnant Services and show that low-productivity-growth Services tend to be relatively income elastic, so that income growth shifts expenditure toward them, while relative-price changes work in the opposite direction. Importantly, they also show that conventional distinctions such as market versus non-market Services do not align closely with productivity-growth differences. Our results point to an additional source of heterogeneity within Services: even conditional on sectoral productivity growth, differences in growth pass-through determine how strongly productivity improvements translate into relative-price changes. Hence, future structural change within Services depends jointly on income elasticities, productivity growth, and growth pass-through.

6.2 Quality Adjustment in Sectoral Prices

A remaining measurement issue is quality adjustment in sectoral prices. To the extent that WIOD prices do not fully capture quality improvements, recovered productivity growth may be biased. This does not affect estimated growth pass-through, however, since $\rho_{j,t}$ depends only on the markup-implied $\kappa_{j,t}$. If unmeasured quality improvement is denoted by $\Delta \log v_{j,t} \geq 0$, such that $\Delta \log P_{j,t}^{true} = \Delta \log P_{j,t}^{WIOD} - \Delta \log v_{j,t}$, then (32) implies

$$\Delta \log a_{j,t}^{*,true} = \Delta \log a_{j,t}^{*,WIOD} + \Delta \log v_{j,t} \tag{34}$$

Hence, unmeasured quality improvement understates recovered cut-off productivity growth.

As a robustness check, I replace WIOD prices with BEA PCE chain-type price indexes. Agri-

culture is mapped to food, Services to PCE Services, and Industry approximately to non-food goods, constructed from the goods and food price indexes and their nominal expenditure shares. The mapping is necessarily imperfect because PCE categories are consumption categories rather than production sectors and measure prices paid by consumers. Since PCE does not distinguish domestic from imported consumption, I compare it with the trade-inclusive WIOD baseline.

Table E.3 compares average annual cut-off productivity growth under the two price sources. The Industry estimate changes little, from 2.39% to 2.61%. For Services, estimated growth falls from 0.62% to 0.26%. Consequently, the Industry-Services productivity-growth gap does not narrow under PCE prices, but instead widens from 1.77 to 2.35 percentage points. The larger discrepancy for Agriculture is of limited aggregate importance given the sector's small expenditure share.

This exercise should be interpreted narrowly. Because the same $\kappa_{j,t}$ path is used under both price measures, growth pass-through and the distributional component of conditional mean productivity are unchanged by construction. Moreover, PCE and WIOD differ in their scope and construction beyond quality adjustment alone. Nevertheless, the fact that the Industry–Services productivity gap survives, and if anything increases, under the alternative price source suggests that insufficient quality adjustment in WIOD is unlikely to be the main explanation for the divergence in recovered productivity growth.

6.3 Endogenous Technology Diffusion

The calibration in Section 4 allows the empirically inferred shape parameter $\kappa_{j,t}$ to vary over time. In the baseline model, productivity growth is exogenous. This section sketches a dynamic extension in which growth is generated endogenously through learning-by-imitation. The shape parameter κ_j is a genuine primitive and shapes both productivity growth and growth pass-through. Rising markups can be explained without exogenous shifts to κ_j, but rather through learning of firms over time. Full derivations are in Appendix A.4.

The static equilibrium of Sections 2.3 closes the model through free entry: a continuum of potential entrants pays a sunk cost F and draw productivity a_{ij} from $\text{Pareto}(A_j, \kappa_j)$, and the cut-off a_j^* equates expected profit to F. To generate ongoing productivity growth, this section introduces learning-by-imitation following Perla and Tonetti (2014). Rather than paying a sunk cost to enter, each sector contains a fixed unit measure of firms. In each period, the least productive share $\chi_j \in (0, 1)$ pauses production to imitate a randomly sampled active producer, re-entering next period with the imitated productivity draw. Because there is no sunk-cost margin to select the cut-

off, the choke price $\bar{p}_{j,t}$ is instead pinned down by labour-market clearing: the sector's employment share $l_{j,t} \equiv L_{j,t}/L$ must equal labour demand across active producers.

Learning-by-imitation truncates the productivity distribution each period: since imitating firms redraw only from the pool of currently active producers, and producing firms are exactly those with $a \geq A_{j,t+1}$, the time t production cut-off becomes the time $t+1$ lower bound of the distribution, $A_{j,t+1} \geq A_{j,t}$. Defining the productivity-frontier growth rate as $g_j \equiv A_{j,t+1}/A_{j,t}$, the truncation implies

$$g_j = (1 - \chi_j)^{-1/\kappa_j} > 1 \tag{35}$$

The growth rate depends only on the imitation share χ_j and the shape parameter κ_j, and is constant over time.

Let $z_{j,t} \equiv \bar{p}_{j,t} A_{j,t+1} \geq 1$ denote the choke price relative to the production cut-off. The sectoral price index can be written as $P_{j,t} = \frac{1}{2}\bar{p}_{j,t}\,\Lambda_{j,t}(\kappa_j, z_{j,t})$, where $\Lambda_{j,t}$ captures the composition of active firms' markups. Then, the growth pass-through is given by:

$$\rho_{j,t} \equiv -\left.\frac{d\log P_{j,t}}{d\log A_{j,t}}\right|_{l_{j,t}} = 1 - \left(1 + \frac{d\log \Lambda_{j,t}}{d\log z_{j,t}}\right)\frac{d\log z_{j,t}}{d\log A_{j,t}} < 1 \tag{36}$$

As in the static model, growth pass-through is incomplete: part of any technology improvement raises $z_{j,t}$ – pushing the choke price further above the marginal producer's marginal cost – rather than being passed into lower prices. Growth pass-through $\rho_{j,t}$ is state-dependent through $z_{j,t}$, hence endogenously time-varying as the frontier moves, without needing κ_j to change. The productivity distribution shapes both the growth rate and the growth pass-through, and does so in opposite directions. Figure 6 illustrates the mechanism: a thicker upper tail makes imitation draws from high-productivity firms more likely, raising the growth rate. At the same time high-productivity firms have the lowest pass-through, so sector-level pass-through falls.

The sectoral markup is given by

$$\mu^c_{j,t} = \frac{\kappa_j + 1}{2\kappa_j} \cdot \frac{(\kappa_j + 2)z^2_{j,t} - \kappa_j}{(\kappa_j + 2)z_{j,t} - (\kappa_j + 1)} \tag{37}$$

At $z_{j,t} \to 1$, (37) collapses to $\mu^c_{j,t} \to 1 + 1/\kappa_j$: the markup formula of Section 2.4 emerges as the limiting case in which the choke price approaches the marginal producer's marginal cost. Away from this limit, $\mu^c_{j,t}$ is increasing in $z_{j,t}$. A rise in $A_{j,t}$ increases $z_{j,t}$, so sectoral markups rise *mechanically* as technology advances, without any change in κ_j. Rising concentration is therefore not necessary to generate rising markups within a sector; it is sufficient for the productivity frontier

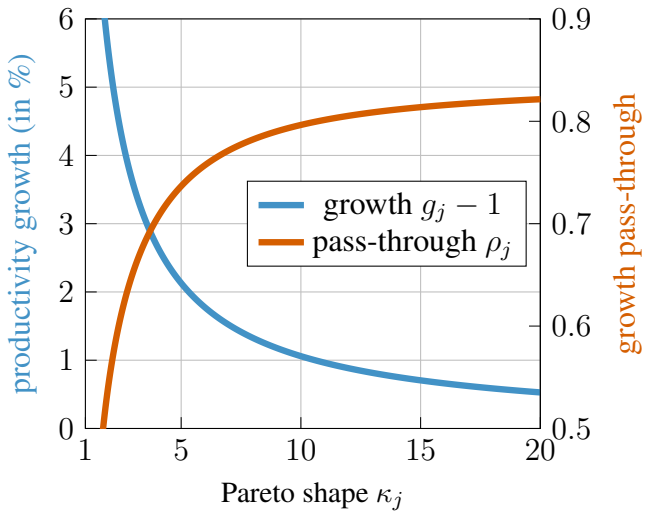


Figure 6: Growth and Pass-Through by Firm Distribution

Notes: The figure illustrates the opposing roles of the Pareto shape parameter. A thicker upper tail raises productivity growth but lowers growth pass-through, and vice versa. The blue line plots the productivity growth rate. The orange line plots the corresponding growth pass-through $\rho(\kappa)$.

to keep moving. This feature is similar to Marto (2024), who achieves rising markups through income-dependent price elasticity of demand.

7 Conclusion

This paper shows that market structure within sectors matters for structural change. Incorporating heterogeneous firms and variable markups into a structural change framework generates incomplete growth pass-through, whose magnitude depends on the productivity distribution. This separates the canonical price effect into underlying productivity growth and its pass-through into the price index. Quantitatively, limited growth pass-through substantially dampens the speed of structural change. In an application to the US from 1980 to 2014, the service expenditure share would have been 2.92 percentage points higher by 2014 in the full pass-through counterfactual. The effect originates primarily in Industry. Over the sample period, growth pass-through has declined in all sectors. Although markup dynamics cannot by themselves explain the slowdown in structural change, they contributed to it by weakening the transmission of productivity growth into sectoral prices. More broadly, the results imply that observed sectoral price changes need not reveal underlying productivity growth one-for-one. Accounting for market structure is therefore important both for understanding structural transformation and for identifying its technological drivers.

References

Acemoglu, Daron and Veronica Guerrieri, "Capital Deepening and Nonbalanced Economic Growth," *Journal of Political Economy*, 2008, *116* (3).

Aghion, Philippe, Antonin Bergeaud, Timo Boppart, Peter J Klenow, and Huiyu Li, "A Theory of Falling Growth and Rising Rents," *Review of Economic Studies*, 2023, *90* (6).

Akcigit, Ufuk, Santiago Caicedo, Ernest Miguelez, Stefanie Stantcheva, and Valerio Sterzi, "Dancing with the Stars: Innovation through Interactions," *NBER Working Paper 24466*, 2018.

Alvarez, Fernando E, Francisco J Buera, and Robert E Lucas, "Models of Idea Flows," *NBER Working Paper 14135*, 2008.

__ , __ , and __ , "Idea Flows, Economic Growth, and Trade," *NBER Working Paper 19667*, 2013.

Amiti, Mary, Oleg Itskhoki, and Jozef Konings, "Importers, Exporters, and Exchange Rate Disconnect," *American Economic Review*, 2014, *104* (7).

__ , __ , and __ , "International Shocks, Variable Markups, and Domestic Prices," *The Review of Economic Studies*, 2019, *86* (6).

Arkolakis, Costas and Monica Morlacco, "Variable Demand Elasticity, Markups, and Pass-Through," 2017. Working Paper.

Auer, Raphael, Ariel Burstein, Sarah Lein, and Jonathan Vogel, "Unequal Expenditure Switching: Evidence from Switzerland," *Review of Economic Studies*, 2024, *91* (5).

Autor, David, David Dorn, Lawrence F Katz, Christina Patterson, and John Van Reenen, "The Fall of the Labor Share and the Rise of Superstar Firms," *The Quarterly Journal of Economics*, 2020, *135* (2).

Baqaee, David R and Ariel Burstein, "Welfare and Output with Income Effects and Taste Shocks," *The Quarterly Journal of Economics*, 2023, *138* (2).

Baqaee, David Rezza and Emmanuel Farhi, "Productivity and Misallocation in General Equilibrium," *The Quarterly Journal of Economics*, 2020, *135* (1).

Barkai, Simcha, "Declining Labor and Capital Shares," *The Journal of Finance*, 2020, *75* (5).

Basu, Susanto, "Are Price-Cost Markups rising in the United States? A Discussion of the Evidence," *Journal of Economic Perspectives*, 2019, *33* (3).

__ and John G Fernald, "Returns to Scale in US Production: Estimates and Implications," *Journal of Political Economy*, 1997, *105* (2).

Baumol, William J, "Macroeconomics of Unbalanced Growth: the Anatomy of Urban Crisis," *American Economic Review*, 1967, *57* (3).

Behrens, Kristian, Giordano Mion, Yasusada Murata, and Jens Suedekum, "Quantifying the Gap between Equilibrium and Optimum under Monopolistic Competition," *The Quarterly Journal of Economics*, 2020, *135* (4).

Benkard, C. Lanier, Ali Yurukoglu, and Anthony Lee Zhang, "Concentration in Product Markets," *American Economic Journal: Microeconomics*, 2026, *18* (2).

Berman, Nicolas, Philippe Martin, and Thierry Mayer, "How do Different Exporters React to Exchange Rate Changes?," *The Quarterly Journal of Economics*, 2012, *127* (1).

Bloom, Nicholas, Mark Schankerman, and John Van Reenen, "Identifying Technology Spillovers and Product Market Rivalry," *Econometrica*, 2013, *81* (4).

Bond, Steve, Arshia Hashemi, Greg Kaplan, and Piotr Zoch, "Some Unpleasant Markup Arithmetic: Production

Function Elasticities and their Estimation from Production Data," *Journal of Monetary Economics*, 2021, *121*.

Boppart, Timo, "Structural Change and the Kaldor Facts in a Growth Model with Relative Price Effects and non-Gorman Preferences," *Econometrica*, 2014, *82* (6).

Buera, Francisco J and Ezra Oberfield, "The Global Diffusion of Ideas," *Econometrica*, 2020, *88* (1).

__ **and Joseph P Kaboski**, "Can Traditional Theories of Structural Change fit the Data?," *Journal of the European Economic Association*, 2009, *7* (2-3).

__ **and** __ , "The Rise of the Service Economy," *American Economic Review*, 2012, *102* (6).

__ **,** __ **, Richard Rogerson, and Juan I Vizcaino**, "Skill-Biased Structural Change," *The Review of Economic Studies*, 2022, *89* (2).

Burstein, Ariel, Javier Cravino, and Marco Rojas, "Input Price Dispersion Across Buyers and Misallocation," *NBER Working Paper 33128*, 2024.

__ **, Vasco M Carvalho, and Basile Grassi**, "Bottom-Up Markup Fluctuations," *The Quarterly Journal of Economics*, 2025.

Campa, Jose Manuel and Linda S Goldberg, "Exchange Rate Pass-Through into Import Prices," *Review of Economics and Statistics*, 2005, *87* (4).

Caron, Justin, Thibault Fally, and James R Markusen, "International Trade Puzzles: A Solution Linking Production and Preferences," *The Quarterly Journal of Economics*, 2014, *129* (3).

Caselli, Francesco, "Accounting for Cross-Country Income Differences," *Handbook of Economic Growth*, 2005, *1*.

Comin, Diego, Danial Lashkari, and Martí Mestieri, "Structural Change with Long-Run Income and Price Effects," *Econometrica*, 2021, *89* (1).

De Loecker, Jan and Frederic Warzynski, "Markups and Firm-Level Export Status," *American Economic Review*, 2012, *102* (6).

__ **, Jan Eeckhout, and Gabriel Unger**, "The Rise of Market Power and the Macroeconomic Implications," *The Quarterly Journal of Economics*, 2020, *135* (2).

__ **, Pinelopi K Goldberg, Amit K Khandelwal, and Nina Pavcnik**, "Prices, Markups, and Trade Reform," *Econometrica*, 2016, *84* (2).

De Ridder, Maarten, "Market Power and Innovation in the Intangible Economy," *American Economic Review*, 2024, *114* (1).

Dhingra, Swati, "Trading Away Wide Brands for Cheap Brands," *American Economic Review*, 2013, *103* (6).

__ **and John Morrow**, "Monopolistic Competition and Optimum Product Diversity under Firm Heterogeneity," *Journal of Political Economy*, 2019, *127* (1).

Dixit, Avinash, "A Model of Duopoly Suggesting a Theory of Entry Barriers," *The Bell Journal of Economics*, 1979.

Duarte, Margarida and Diego Restuccia, "The Role of the Structural Transformation in Aggregate Productivity," *The Quarterly Journal of Economics*, 2010, *125* (1).

Duernecker, Georg, Berthold Herrendorf, and Akos Valentinyi, "Structural Change within the Services Sector and the Future of Cost Disease," *Journal of the European Economic Association*, 2024, *22* (1).

Eckel, Carsten and J Peter Neary, "Multi-Product Firms and Flexible Manufacturing in the Global Economy," *The Review of Economic Studies*, 2010, *77* (1).

Edmond, Chris, Virgiliu Midrigan, and Daniel Yi Xu, "How Costly are Markups?," *Journal of Political Economy*, 2023, *131* (7).

Fally, Thibault, "Generalized Separability and Integrability: Consumer Demand with a Price Aggregator," *Journal of*

Economic Theory, 2022, *203*.

Feenstra, Robert C, Robert Inklaar, and Marcel P Timmer, "The Next Generation of the Penn World Table," *American Economic Review*, 2015, *105* (10).

Föllmi, Reto and Josef Zweimüller, "Income Distribution and Demand-Induced Innovations," *The Review of Economic Studies*, 2006, *73* (4).

__ **and** __ , "Structural Change, Engel's Consumption Cycles and Kaldor's Facts of Economic Growth," *Journal of Monetary Economics*, 2008, *55* (7).

Gopinath, Gita and Roberto Rigobon, "Sticky Borders," *The Quarterly Journal of Economics*, 2008, *123* (2).

__ **, Oleg Itskhoki, and Roberto Rigobon**, "Currency Choice and Exchange Rate Pass-Through," *American Economic Review*, 2010, *100* (1).

Grullon, Gustavo, Yelena Larkin, and Roni Michaely, "Are US Industries becoming more Concentrated?," *Review of Finance*, 2019, *23* (4).

Gutiérrez, Germán and Thomas Philippon, "Investment-Less Growth: An Empirical Investigation," *NBER Working Paper 22897*, 2016.

Hall, Robert E, "The Relation between Price and Marginal Cost in US Industry," *Journal of Political Economy*, 1988, *96* (5).

Herrendorf, Berthold, Christopher Herrington, and Akos Valentinyi, "Sectoral Technology and Structural Transformation," *American Economic Journal: Macroeconomics*, 2015, *7* (4).

__ **, Richard Rogerson, and Akos Valentinyi**, "Two Perspectives on Preferences and Structural Transformation," *American Economic Review*, 2013, *103* (7).

__ **,** __ **, and** __ , "Structural Change in Investment and Consumption—A Unified Analysis," *The Review of Economic Studies*, 2021, *88* (3).

Klette, Tor Jakob, "Market Power, Scale Economies and Productivity: Estimates from a Panel of Establishment Data," *The Journal of Industrial Economics*, 1999, *47* (4).

Kongsamut, Piyabha, Sergio Rebelo, and Danyang Xie, "Beyond Balanced Growth," *The Review of Economic Studies*, 2001, *68* (4).

Kortum, Samuel S, "Research, Patenting, and Technological Change," *Econometrica*, 1997.

Kwon, Spencer Y, Yueran Ma, and Kaspar Zimmermann, "100 Years of Rising Corporate Concentration," *American Economic Review*, 2024, *114* (7).

Lashkari, Danial, Arthur Bauer, and Jocelyn Boussard, "Information Technology and Returns to Scale," *American Economic Review*, 2024, *114* (6).

Lucas, Robert E and Benjamin Moll, "Knowledge Growth and the Allocation of Time," *Journal of Political Economy*, 2014, *122* (1).

Marto, Ricardo, "Structural Change and the Rise in Markups," 2024. Working Paper.

Matsuyama, Kiminori, "The Rise of Mass Consumption Societies," *Journal of Political Economy*, 2002, *110* (5).

__ , "Engel's Law in the Global Economy: Demand-Induced Patterns of Structural Change, Innovation, and Trade," *Econometrica*, 2019, *87* (2).

Mayer, Thierry, Marc J Melitz, and Gianmarco Ottaviano, "Market Size, Competition, and the Product Mix of Exporters," *American Economic Review*, 2014, *104* (2).

Melitz, Marc J and Gianmarco Ottaviano, "Market Size, Trade, and Productivity," *The Review of Economic Studies*, 2008, *75* (1).

Moretti, Enrico, "The Effect of High-Tech Clusters on the Productivity of Top Inventors," *American Economic Review*, 2021, *111* (10).

Mrázová, Monika and J Peter Neary, "Not so demanding: Demand structure and firm behavior," *American Economic Review*, 2017, *107* (12).

Murphy, Kevin M, Andrei Shleifer, and Robert Vishny, "Income Distribution, Market Size, and Industrialization," *The Quarterly Journal of Economics*, 1989, *104* (3).

Nakamura, Emi, "Pass-Through in Retail and Wholesale," *American Economic Review*, 2008, *98* (2).

__ **and Dawit Zerom**, "Accounting for Incomplete Pass-Through," *The Review of Economic Studies*, 2010, *77* (3).

Ngai, L Rachel and Christopher A Pissarides, "Structural Change in a Multisector Model of Growth," *American Economic Review*, 2007, *97* (1).

Ottaviano, Gianmarco IP and Davide Suverato, "Fantastic Beasts and where to Find Them," 2024. Working Paper.

Parenti, Mathieu, Philip Ushchev, and Jacques-François Thisse, "Toward a Theory of Monopolistic Competition," *Journal of Economic theory*, 2017, *167*.

Perla, Jesse and Christopher Tonetti, "Equilibrium Imitation and Growth," *Journal of Political Economy*, 2014, *122* (1).

Pollak, Robert A, "Additive Utility Functions and Linear Engel Curves," *The Review of Economic Studies*, 1971, *38* (4).

Raval, Devesh, "Testing the Production Approach to Markup Estimation," *Review of Economic Studies*, 2023, *90* (5).

Redding, Stephen J and David E Weinstein, "Measuring Aggregate Price Indices with Taste Shocks: Theory and Evidence for CES Preferences," *The Quarterly Journal of Economics*, 2020, *135* (1).

Ridder, Maarten De, Basile Grassi, and Giovanni Morzenti, "The Hitchhiker's Guide to Markup Estimation: Assessing Estimates from Financial Data," *Econometrica*, 2026, *94* (1).

Timmer, Marcel P, Erik Dietzenbacher, Bart Los, Robert Stehrer, and Gaaitzen J De Vries, "An Illustrated User Guide to the World Input–Output Database: The Case of Global Automotive Production," *Review of International Economics*, 2015, *23* (3).

Traina, James, "Is aggregate market power increasing? Production Trends using Financial Statements," *Working Paper*, 2018.

Wharton Research Data Services, "." 2026. https://wrds-www.wharton.upenn.edu/.

World Input-Output Database, "." 2026. https://www.rug.nl/ggdc/valuechain/wiod/.

Zhelobodko, Evgeny, Sergey Kokovin, Mathieu Parenti, and Jacques-François Thisse, "Monopolistic Competition: Beyond the Constant Elasticity of Substitution," *Econometrica*, 2012, *80* (6).

Appendix

A Theory

Sector indices are suppress whenever possible without confusion.

A.1 Demand and Preferences

Sectoral Demand and Expenditure Shares. Consider the non-homothetic CES aggregator across sectors

$$\sum_{j\in\mathcal{J}} \beta_j^{\frac{1}{\eta}} Q_j^{\frac{\eta-1}{\eta}} Q^{\frac{\epsilon_j-\eta}{\eta}} = 1 \tag{38}$$

with sectoral prices $\{P_j\}_{j\in\mathcal{J}}$ and total expenditure $E = \sum_{j\in\mathcal{J}} P_j Q_j$ for a given consumer. The consumer chooses $\{Q_j\}_{j\in\mathcal{J}}$ and the aggregate real consumption index Q to solve

$$\max_{Q,\{Q_j\}} \; Q + \lambda\left(1 - \sum_{j\in\mathcal{J}} \beta_j^{\frac{1}{\eta}} Q_j^{\frac{\eta-1}{\eta}} Q^{\frac{\epsilon_j-\eta}{\eta}}\right) + \mu\left(E - \sum_{j\in\mathcal{J}} P_j Q_j\right).$$

The first-order condition with respect to Q_j is

$$\frac{\partial \mathcal{L}}{\partial Q_j} = \lambda \frac{1-\eta}{\eta} \beta_j^{\frac{1}{\eta}} Q_j^{-\frac{1}{\eta}} Q^{\frac{\epsilon_j-\eta}{\eta}} - \mu P_j = 0.$$

Rearranging and multiplying by Q_j gives

$$\lambda \frac{1-\eta}{\eta} \beta_j^{\frac{1}{\eta}} Q_j^{\frac{\eta-1}{\eta}} Q^{\frac{\epsilon_j-\eta}{\eta}} = \mu P_j Q_j.$$

Summing over all sectors and using (38), we obtain

$$\lambda \frac{1-\eta}{\eta} \sum_{j\in\mathcal{J}} \beta_j^{\frac{1}{\eta}} Q_j^{\frac{\eta-1}{\eta}} Q^{\frac{\epsilon_j-\eta}{\eta}} = \mu \sum_{j\in\mathcal{J}} P_j Q_j \quad \Longrightarrow \quad \lambda \frac{1-\eta}{\eta} = \mu E.$$

Substituting this back into the first-order condition and solving for Q_j delivers sectoral demand

$$Q_j = \beta_j E^{\eta} P_j^{-\eta} Q^{\epsilon_j-\eta} = \beta_j \left(\frac{P_j}{P}\right)^{-\eta} Q^{\epsilon_j}, \tag{39}$$

where we used $E = PQ$. The aggregate price index is then pinned down by the identity $E = PQ = \sum_{j\in\mathcal{J}} P_j Q_j$. Substituting (39) into this expression gives

$$P = \left(\sum_{j\in\mathcal{J}} \beta_j P_j^{1-\eta} Q^{\epsilon_j-1}\right)^{\frac{1}{1-\eta}}.$$

Finally, the expenditure share of sector j is

$$\omega_j = \beta_j \left(\frac{P_j}{P}\right)^{1-\eta} Q^{\epsilon_j - 1}.$$

Individual Demand. Using utility (7) and the budget constraint (8), the Lagrangian reads:

$$\mathcal{L} = \int_{i\in\Omega} \alpha q_i - \frac{\gamma}{2} q_i^2 \ di - \lambda \left(\int_{i\in\Omega} p_i q_i \ di - E \right)$$

The FOC with respect to q_i is:

$$\frac{\partial \mathcal{L}}{\partial q_i} : \alpha - \gamma q_i = \lambda p_i \quad \Leftrightarrow \quad q_i = \frac{\alpha - \lambda p_i}{\gamma}$$

The choke price $\bar{p}$ is by setting demand equal to zero, thus we have $\bar{p} = \alpha/\lambda$. Plugging this back into the FOC, we have:

$$q_i = \frac{\lambda(\bar{p} - p_i)}{\gamma} = \frac{\bar{p}\lambda}{\gamma}\frac{(\bar{p} - p_i)}{\bar{p}} = \frac{\alpha}{\gamma}\left(1 - \frac{p_i}{\bar{p}}\right)$$

which matches (9).

Derive Pass-through. Let $x = p/\bar{p}$ be the relative price. Suppose the firm sets price as a markup over marginal cost $p = \mu(x)\, c$, where the markup $\mu(x)$ depends only on x. Define the markup elasticity and the pass-through as

$$\Gamma(x) \equiv -\frac{d\log \mu(x)}{d\log x}, \qquad \Phi(x) \equiv \frac{d\log p}{d\log c}$$

Because $x = p/\bar{p}$, we have $x\bar{p} = \mu(x)c$. Taking logs,

$$\log x + \log \bar{p} = \log \mu(x) + \log c \quad \Rightarrow \quad \log x = \log c + \log \mu(x) - \log \bar{p}.$$

Differentiating with respect to $\log c$ and using the chain rule,

$$\begin{aligned} \frac{d\log x}{d\log c} &= 1 + \frac{d\log \mu}{d\log c} = 1 + \frac{d\log \mu}{d\log x}\frac{d\log x}{d\log c} \\ &= 1 - \Gamma(x)\frac{d\log x}{d\log c}. \end{aligned}$$

Hence

$$\big(1 + \Gamma(x)\big)\frac{d\log x}{d\log c} = 1 \quad \Leftrightarrow \quad \frac{d\log x}{d\log c} = \frac{1}{1+\Gamma(x)}$$

Finally, since $p = x\bar{p}$ and $\bar{p}$ is constant,

$$\Phi(x) = \frac{d\log p}{d\log c} = \frac{d\log x}{d\log c} = \frac{1}{1+\Gamma(x)}$$

For the quadratic demand system considered in the main text, the markup is given by $\mu(x) = (2 - 1/x)^{-1}$. The

corresponding markup elasticity is

$$\Gamma(x) = -\frac{d\log\mu(x)}{d\log x} = \frac{1}{2x-1} = \frac{\varepsilon+1}{\varepsilon-1} > 0, \qquad \Gamma'(x) = -\frac{2}{(2x-1)^2} < 0. \tag{40}$$

Substituting into the pass-through expression yields

$$\Phi(x) = \frac{1}{1+\Gamma(x)} = \frac{2x-1}{2x}, \tag{41}$$

which coincides with the expression derived directly from the firm's pricing rule in the main text. Given that $\Gamma(x) > 0$, we have incomplete pass-through: changes in marginal costs are not fully passed on to consumers, but partly absorbed through markup adjustments. Since $\Gamma'(x) < 0$, pass-through increases with the relative price x. As more productive firms charge lower prices relative to the choke price, they exhibit lower pass-through. Hence, large, high-productivity firms have lower pass-through than low-productivity firms operating close to the choke price.

A.2 Equilibrium

This section derives the expressions for sectoral expenditure E_j, the sectoral quantity index Q_j, and the corresponding price index $P_j = E_j/Q_j$ used in the main text.

Expenditure. Sectoral expenditure per consumer is

$$E_j = \int_0^{N_j} p_{ij}q_{ij}di = N_j \int_{a_j^*}^{\infty} p_j(a)q_j(a)\, g_j(a \mid a \geq a_j^*)\, da.$$

Using the equilibrium price and individual demand schedules $p_j(a) = \frac{1}{2}\left(a^{-1} + (a_j^*)^{-1}\right)$ and $q_j(a) = \frac{\alpha}{2\gamma}\left(1 - \frac{a_j^*}{a}\right)$, sectoral expenditure can be written as

$$\begin{aligned} E_j &= \frac{\alpha N_j}{4\gamma}\left[\frac{1}{a_j^*} - a_j^* \int_{a_j^*}^{\infty} a^{-2} g_j(a \mid a \geq a_j^*)\, da\right] \\ &= \frac{\alpha N_j}{4\gamma a_j^*}\left(1 - \frac{\kappa_j}{\kappa_j+2}\right) = \frac{\alpha N_j}{2\gamma a_j^*(\kappa_j+2)}. \end{aligned}$$

Quantity. Sectoral real consumption is defined as the total physical quantity consumed across producing varieties,

$$Q_j = \int_0^{N_j} q_{ij}di = N_j \int_{a_j^*}^{\infty} q_j(a)\, g_j(a \mid a \geq a_j^*)\, da.$$

Using $\int_{a_j^*}^{\infty} a^{-1} g_j(a \mid a \geq a_j^*)\, da = \frac{\kappa_j}{\kappa_j+1}(a_j^*)^{-1}$ gives

$$\begin{aligned} Q_j &= \frac{\alpha N_j}{2\gamma}\left[1 - a_j^* \int_{a_j^*}^{\infty} a^{-1} g_j(a \mid a \geq a_j^*)\, da\right] \\ &= \frac{\alpha N_j}{2\gamma}\left(1 - \frac{\kappa_j}{\kappa_j+1}\right) = \frac{\alpha N_j}{2\gamma(\kappa_j+1)}. \end{aligned}$$

Sectoral Price Index. Finally, the sectoral unit-value price index is defined as $P_j \equiv E_j/Q_j$. Combining the two expressions above yields

$$P_j = \frac{\kappa_j + 1}{\kappa_j + 2} \frac{1}{a_j^*}. \tag{42}$$

Substituting the equilibrium productivity cut-off a_j^* gives

$$P_j = \frac{\kappa_j + 1}{\kappa_j + 2} \left(\frac{F}{L\delta_\pi(\kappa_j)} \right)^{\frac{1}{1+\kappa_j}} A_j^{-\frac{\kappa_j}{1+\kappa_j}}, \tag{43}$$

which is the expression used in the main text.

A.3 Analytical Results

Cost-Weighted Markups. Recall that the cost-weighted sectoral markup is defined as

$$\mu_j^c \equiv \frac{\int_{a_j^*}^\infty \mu_j(a) a^{-1} y_j(a)\, dG_j(a)}{\int_{a_j^*}^\infty a^{-1} y_j(a)\, dG_j(a)} = \frac{\int_{a_j^*}^\infty p_j(a) y_j(a)\, dG_j(a)}{\int_{a_j^*}^\infty a^{-1} y_j(a)\, dG_j(a)}$$

where the second equality follows from $\mu_j(a)a^{-1} = p_j(a)$. Using the equilibrium price and quantity schedules $p_j(a) = \frac{1}{2}\left(a^{-1} + (a_j^*)^{-1}\right)$ and $y_j(a) = L\frac{\alpha}{2\gamma}\left(1 - \frac{a_j^*}{a}\right)$, together with the Pareto density $g_j(a) = \kappa_j A_j^{\kappa_j} a^{-\kappa_j - 1}$, sectoral revenue is

$$\begin{aligned} \int_{a_j^*}^\infty p_j(a) y_j(a)\, dG_j(a) &= \frac{L\alpha}{4\gamma} \left[\frac{1}{a_j^*} \int_{a_j^*}^\infty dG_j(a) - a_j^* \int_{a_j^*}^\infty a^{-2}\, dG_j(a) \right] \\ &= \frac{L\alpha}{2\gamma} \frac{A_j^{\kappa_j}}{(a_j^*)^{\kappa_j+1}(\kappa_j+2)}. \end{aligned}$$

Likewise, sectoral variable cost is

$$\begin{aligned} \int_{a_j^*}^\infty a^{-1} y_j(a)\, dG_j(a) &= \frac{L\alpha}{2\gamma} \left[\int_{a_j^*}^\infty a^{-1}\, dG_j(a) - a_j^* \int_{a_j^*}^\infty a^{-2}\, dG_j(a) \right] \\ &= \frac{L\alpha}{2\gamma} \frac{\kappa_j A_j^{\kappa_j}}{(a_j^*)^{\kappa_j+1}(\kappa_j+1)(\kappa_j+2)}. \end{aligned}$$

Taking the ratio gives

$$\mu_j^c = \frac{\kappa_j + 1}{\kappa_j} = 1 + \frac{1}{\kappa_j},$$

which is equation (21) in the main text.

Sales-Weighted Markups. For comparison to the cost-weighted markups derived in the main text, consider now the sales-weighted sectoral markup, which weights firms by their revenue shares rather than their cost shares. In sector j,

define

$$\mu_j^R \equiv \frac{\int_{a_j^*}^{\infty} \mu_{ij}(a)\, p_{ij}(a)\, y_{ij}(a)\, dG_j(a)}{\int_{a_j^*}^{\infty} p_{ij}(a)\, y_{ij}(a)\, dG_j(a)},$$

so the weight of firm i is its share in total sales $p_{ij}y_{ij}$. Using the equilibrium price and quantity schedules, $p_{ij}(a) = \frac{1}{2}\big(a^{-1} + (a_j^*)^{-1}\big)$ and $y_{ij}(a) = L\frac{\alpha}{2\gamma}\big(1 - a_j^*/a\big)$, together with the Pareto density $g_j(a) = \kappa_j A_j^{\kappa_j} a^{-\kappa_j - 1}$, we obtain

$$\int_{a_j^*}^{\infty} p_{ij}(a)\, q_{ij}(a)\, g_j(a)\, da = \frac{\alpha}{2\gamma}\, \frac{A_j^{\kappa_j}}{(a_j^*)^{\kappa_j+1}(\kappa_j + 2)},$$

$$\int_{a_j^*}^{\infty} \mu_{ij}(a)\, p_{ij}(a)\, q_{ij}(a)\, g_j(a)\, da = \frac{\alpha}{2\gamma}\, \frac{A_j^{\kappa_j}}{(a_j^*)^{\kappa_j+1}}\, \frac{2\kappa_j^2 + 2\kappa_j - 1}{2(\kappa_j + 2)(\kappa_j^2 - 1)}.$$

Dividing the two expressions yields the sales-weighted sectoral markup

$$\mu_j^R = \frac{\kappa_j^2 + \kappa_j - \frac{1}{2}}{\kappa_j^2 - 1} = \left(1 + \frac{1}{\kappa_j}\right) + \frac{\kappa_j + 2}{2\kappa_j(\kappa_j^2 - 1)} = \mu_j^c + \frac{\kappa_j + 2}{2\kappa_j(\kappa_j^2 - 1)} > \mu_j^c \tag{44}$$

Sales weights put more emphasis on high-productivity, high-markup firms, so μ_j^R exceeds the cost-weighted markup μ_j^c in every sector. Both measures fall with κ_j and converge to one as the productivity distribution becomes more concentrated near the lower bound (i.e. as $\kappa_j \to \infty$). As $\kappa_j \to 1$ the sales-weighted markup diverges while the cost-weighted markup remains finite at $\mu_j^c \to 2$, reflecting the increasing dominance of extremely productive, high-markup firms in sectoral sales.

Proof of Proposition 2: Growth & Structural Change. Recall the expenditure share implied by the non-homothetic CES aggregator:

$$\begin{aligned} \omega_k &= \beta_k \left(\frac{P_k}{P}\right)^{1-\eta} Q^{\epsilon_k - 1}, \\ \Leftrightarrow \qquad \log \omega_k &= \log \beta_k + (1-\eta)\big(\log P_k - \log P\big) + (\epsilon_k - 1)\log Q, \end{aligned}$$

where $E = PQ$ and $\bar{\epsilon} \equiv \sum_{m \in \mathcal{J}} \omega_m \epsilon_m$ denotes the expenditure-weighted average income elasticity.

Consider a sector-specific technology improvement in sector j, modelled as an increase in A_j, holding A_k fixed for all $k \neq j$. Differentiating $\log \omega_k$ with respect to $\log A_j$ gives, for any k,

$$\frac{d\log \omega_k}{d\log A_j} = (1-\eta)\left(\frac{d\log P_k}{d\log A_j} - \frac{d\log P}{d\log A_j}\right) + (\epsilon_k - 1)\frac{d\log Q}{d\log A_j}. \tag{45}$$

From the aggregate price deflator (4), total differentiation yields

$$(1-\eta)\, d\log P = (1-\eta) \sum_{m \in \mathcal{J}} \omega_m\, d\log P_m + (\bar{\epsilon} - 1)\, d\log Q. \tag{46}$$

Equivalently,

$$d\log P = \sum_{m \in \mathcal{J}} \omega_m\, d\log P_m + \frac{\bar{\epsilon} - 1}{1-\eta}\, d\log Q. \tag{47}$$

Since the sectoral price index P_k depends on the technology level A_k of sector k, a technology change in sector j leaves the price indices of all other sectors unchanged. Hence $\frac{d\log P_k}{d\log A_j} = 0$ for all $k \neq j$, while Proposition 1 implies $\frac{d\log P_j}{d\log A_j} = -\rho_j$. It follows that

$$\sum_{m\in\mathcal{J}} \omega_m \frac{d\log P_m}{d\log A_j} = -\omega_j\rho_j. \tag{48}$$

Using (47), we therefore obtain

$$\frac{d\log P}{d\log A_j} = -\omega_j\rho_j + \frac{\bar{\epsilon}-1}{1-\eta}\frac{d\log Q}{d\log A_j}. \tag{49}$$

Under free entry, expected net profits are zero and nominal expenditure per consumer is fixed at $E = 1$. Since $E = PQ$, this implies $d\log Q = -d\log P$. Combining this result with (49) gives

$$-\frac{d\log Q}{d\log A_j} = -\omega_j\rho_j + \frac{\bar{\epsilon}-1}{1-\eta}\frac{d\log Q}{d\log A_j},$$

and therefore

$$\frac{d\log Q}{d\log A_j} = \frac{(1-\eta)\omega_j\rho_j}{\bar{\epsilon}-\eta} > 0, \tag{50}$$

where the inequality follows from $\bar{\epsilon} > \eta$.

Now plug (49) into (45). First consider $k = j$. Using $\frac{d\log P_j}{d\log A_j} = -\rho_j$, we obtain

$$\begin{aligned}\frac{d\log\omega_j}{d\log A_j} &= (1-\eta)\left[-\rho_j + \omega_j\rho_j - \frac{\bar{\epsilon}-1}{1-\eta}\frac{d\log Q}{d\log A_j}\right] + (\epsilon_j - 1)\frac{d\log Q}{d\log A_j} \\ &= -(1-\eta)(1-\omega_j)\rho_j + (\epsilon_j - \bar{\epsilon})\frac{d\log Q}{d\log A_j}.\end{aligned} \tag{51}$$

Substituting (50) into (51) yields

$$\frac{d\log\omega_j}{d\log A_j} = -(1-\eta)\rho_j\left[1 - \frac{\omega_j(\epsilon_j-\eta)}{\bar{\epsilon}-\eta}\right], \tag{52}$$

which gives (24).

Next consider any sector $k \neq j$. Since $\frac{d\log P_k}{d\log A_j} = 0$, the same substitutions yield

$$\begin{aligned}\frac{d\log\omega_k}{d\log A_j} &= (1-\eta)\left[\omega_j\rho_j - \frac{\bar{\epsilon}-1}{1-\eta}\frac{d\log Q}{d\log A_j}\right] + (\epsilon_k - 1)\frac{d\log Q}{d\log A_j} \\ &= (1-\eta)\omega_j\rho_j + (\epsilon_k - \bar{\epsilon})\frac{d\log Q}{d\log A_j} \\ &= (1-\eta)\omega_j\rho_j\frac{\epsilon_k-\eta}{\bar{\epsilon}-\eta}, \qquad k \neq j.\end{aligned} \tag{53}$$

This gives (25).

Since sectoral employment satisfies $L_k = L\omega_k$ and aggregate labour supply L is fixed, expenditure shares and

employment have identical elasticities:

$$\frac{d\log L_k}{d\log A_j} = \frac{d\log \omega_k}{d\log A_j}, \qquad \forall k \in \mathcal{J}. \tag{54}$$

Finally, consider the number of producing firms. Recall that

$$N_k = \frac{L_k}{F(1+\kappa_k)}\left(\frac{A_k}{a_k^*}\right)^{\kappa_k}.$$

For any non-innovating sector $k \neq j$, both A_k and a_k^* are unaffected by the technology change in sector j. Hence

$$\frac{d\log N_k}{d\log A_j} = \frac{d\log L_k}{d\log A_j} = \frac{d\log \omega_k}{d\log A_j}, \qquad k \neq j. \tag{55}$$

For the innovating sector, differentiating the expression for N_j gives

$$\frac{d\log N_j}{d\log A_j} = \frac{d\log L_j}{d\log A_j} + \kappa_j\left(1 - \frac{d\log a_j^*}{d\log A_j}\right).$$

From the equilibrium cut-off, $\frac{d\log a_j^*}{d\log A_j} = \frac{\kappa_j}{1+\kappa_j} = \rho_j$, so $\kappa_j(1-\rho_j) = \rho_j$. Therefore,

$$\begin{aligned}\frac{d\log N_j}{d\log A_j} &= \frac{d\log \omega_j}{d\log A_j} + \rho_j \\ &= \rho_j\left[\eta + (1-\eta)\frac{\omega_j(\epsilon_j - \eta)}{\bar{\epsilon} - \eta}\right],\end{aligned} \tag{56}$$

which gives (26). This completes the proof. ■

A.4 Endogenous Technology Diffusion

Truncation & Law of Motion. Let $G_{j,t}(a)$ denote the distribution of productivity draws across the unit mass of potential firms in sector j at date t, with support $[A_{j,t}, \infty)$. A constant share $\chi_j \in (0,1)$ of firms is engaged in learning-by-imitation and does not produce, so the mass of producing firms is $1-\chi_j$. Denote by $A_{j,t+1} \geq A_{j,t}$ the productivity cut-off of the marginal active producer. Since firms produce if and only if $a \geq A_{j,t+1}$, this implies

$$1 - \chi_j = 1 - G_{j,t}(A_{j,t+1}). \tag{57}$$

Imitating firms redraw from the cross-section of active producers, so the relevant sampling distribution is the distribution conditional on producing,

$$G_{j,t}^P(a) \equiv \Pr(\tilde{a} \leq a \mid \tilde{a} \geq A_{j,t+1}) = \frac{G_{j,t}(a) - G_{j,t}(A_{j,t+1})}{1 - G_{j,t}(A_{j,t+1})}, \qquad a \in [A_{j,t+1}, \infty),$$

with density $g_{j,t}^P(a) = g_{j,t}(a)/(1 - G_{j,t}(A_{j,t+1}))$ on $[A_{j,t+1}, \infty)$. This implies that the productivity distribution is truncated at the producer cut-off: after imitation and updating, the next period density is a renormalized truncation of

the current density,

$$g_{j,t+1}(a) = \frac{g_{j,t}(a)}{1 - G_{j,t}(A_{j,t+1})}, \qquad a \in [A_{j,t+1}, \infty), \tag{58}$$

and zero otherwise. Iterating (58) forward, the cross-sectional distribution at any date is fully characterized by the initial distribution and the current truncation point: the density remains Pareto with shape κ_j and lower bound $A_{j,t}$ at every date t. Using the Pareto distribution, (57), and defining the growth rate as the change in the productivity frontier $g_j \equiv A_{j,t+1}/A_{j,t}$, we get

$$1 - \chi_j = \left(\frac{A_{j,t}}{A_{j,t+1}}\right)^{\kappa_j} = g_j^{-\kappa_j} \quad \Longleftrightarrow \quad g_j = (1-\chi_j)^{-1/\kappa_j} > 1, \tag{59}$$

which is the growth rate equation stated in the main text. The imitation share χ_j pins down the rate at which the productivity frontier expands over time: a larger share of imitators implies a faster expansion, while a larger shape parameter κ_j lowers the growth rate for a given imitation share, because the productivity distribution is more concentrated near its lower bound. Combining (59) with the definition of g_j yields the law of motion for the productivity frontier:

$$A_{j,t+1} = g_j A_{j,t} = (1-\chi_j)^{-1/\kappa_j} A_{j,t}$$

Equilibrium. At any date t, an active firm with productivity $a \geq A_{j,t+1}$ sets price and quantity

$$p(a) = \tfrac{1}{2}\left(\bar{p}_{j,t} + \tfrac{1}{a}\right), \qquad q(a) = \frac{\alpha}{2\gamma}\left(1 - \frac{1}{a\bar{p}_{j,t}}\right).$$

Since the mass of firms in sector j is fixed at one, sectoral expenditure and real consumption are obtained by integrating the unconditional Pareto density over the active region $[A_{j,t+1}, \infty)$:

$$E_{j,t} = \int_{A_{j,t+1}}^{\infty} p(a)q(a)\, g_{j,t}(a)\, da, \qquad Q_{j,t} = \int_{A_{j,t+1}}^{\infty} q(a)\, g_{j,t}(a)\, da.$$

The two integrands yield.

$$\begin{aligned} p(a)q(a) &= \frac{\alpha}{4\gamma}\left(\bar{p}_{j,t} + \frac{1}{a}\right)\left(1 - \frac{1}{a\bar{p}_{j,t}}\right) = \frac{\alpha}{4\gamma}\left[\bar{p}_{j,t} - \frac{1}{a^2\bar{p}_{j,t}}\right], \\ q(a) &= \frac{\alpha}{2\gamma}\left[1 - \frac{1}{a\bar{p}_{j,t}}\right] \end{aligned}$$

The three Pareto moments thus required satisfy:

$$\begin{aligned} \int_{A_{j,t+1}}^{\infty} g_{j,t}(a)\, da &= \left(\frac{A_{j,t}}{A_{j,t+1}}\right)^{\kappa_j} = 1 - G_{j,t}(A_{j,t+1}), \\ \int_{A_{j,t+1}}^{\infty} a^{-1} g_{j,t}(a)\, da &= \frac{\kappa_j}{\kappa_j + 1}\big(1 - G_{j,t}(A_{j,t+1})\big) A_{j,t+1}^{-1}, \\ \int_{A_{j,t+1}}^{\infty} a^{-2} g_{j,t}(a)\, da &= \frac{\kappa_j}{\kappa_j + 2}\big(1 - G_{j,t}(A_{j,t+1})\big) A_{j,t+1}^{-2}, \end{aligned}$$

Substituting these into the definitions of $Q_{j,t}$ and $E_{j,t}$ gives:

$$E_{j,t} = \frac{\alpha \bar{p}_{j,t}\big(1 - G_{j,t}(A_{j,t+1})\big)}{4\gamma}\left[1 - \frac{\kappa_j}{\kappa_j + 2}\big(A_{j,t+1}\bar{p}_{j,t}\big)^{-2}\right], \tag{60}$$

$$Q_{j,t} = \frac{\alpha\big(1 - G_{j,t}(A_{j,t+1})\big)}{2\gamma}\left[1 - \frac{\kappa_j}{\kappa_j + 1}\big(A_{j,t+1}\bar{p}_{j,t}\big)^{-1}\right] \tag{61}$$

Dividing (60) by (61) and defining $z_{j,t} = A_{j,t+1}\bar{p}_{j,t}$ leaves

$$P_{j,t} \equiv \frac{E_{j,t}}{Q_{j,t}} = \frac{\bar{p}_{j,t}}{2}\frac{1 - \frac{\kappa_j}{\kappa_j+2}(A_{j,t+1}\bar{p}_{j,t})^{-2}}{1 - \frac{\kappa_j}{\kappa_j+1}(A_{j,t+1}\bar{p}_{j,t})^{-1}} = \frac{\bar{p}_{j,t}}{2}\frac{1 - \frac{\kappa_j}{\kappa_j+2}(z_{j,t})^{-2}}{1 - \frac{\kappa_j}{\kappa_j+1}(z_{j,t})^{-1}} = \frac{\bar{p}_{j,t}}{2}\Lambda_{j,t}(\kappa_j, z_{j,t}) \tag{62}$$

which is the price-index decomposition used in the main text.

Labour market clearing requires that labour demand across active producers, $\int_{A_{j,t+1}}^{\infty} a^{-1} q(a)\, dG_{j,t}(a)$, equal the sector's employment share $l_{j,t}$:

$$l_{j,t} = \frac{\alpha\big(1 - G_{j,t}(A_{j,t+1})\big)}{2\gamma}\left[\frac{\kappa_j}{\kappa_j + 1}A_{j,t+1}^{-1} - \frac{1}{\bar{p}_{j,t}} \cdot \frac{\kappa_j}{\kappa_j + 2}A_{j,t+1}^{-2}\right] \tag{63}$$

Rearranging (63) allows us to solve explicitly for the equilibrium choke price $\bar{p}_{j,t}$. Starting from the labour market clearing condition, define $B_j \equiv \frac{\alpha\kappa_j(1-\chi_j)}{2\gamma}$. Then the two terms in (63) simplify to

$$\frac{\alpha}{2\gamma}\frac{\kappa_j}{\kappa_j + 1}(1 - \chi_j)A_{j,t+1}^{-1} = \frac{B_j}{\kappa_j + 1}A_{j,t+1}^{-1}, \qquad \frac{\alpha}{2\gamma}\frac{\kappa_j}{\kappa_j + 2}(1 - \chi_j)A_{j,t+1}^{-2} = \frac{B_j A_{j,t+1}^{-2}}{\kappa_j + 2}.$$

Substituting into (63) yields

$$l_{j,t} = \frac{B_j A_{j,t+1}^{-1}}{\kappa_j + 1} - \frac{1}{\bar{p}_{j,t}}\frac{B_j A_{j,t+1}^{-2}}{\kappa_j + 2} \tag{64}$$

Rearranging (64) gives

$$\frac{1}{\bar{p}_{j,t}}\frac{B_j A_{j,t+1}^{-2}}{\kappa_j + 2} = \frac{B_j A_{j,t+1}^{-1}}{\kappa_j + 1} - l_{j,t} \quad \Longrightarrow \quad \frac{1}{\bar{p}_{j,t}} = \frac{\kappa_j + 2}{B_j A_{j,t+1}^{-2}}\left(\frac{B_j A_{j,t+1}^{-1}}{\kappa_j + 1} - l_{j,t}\right)$$

Inverting and simplifying implies the closed form for the choke price:

$$\bar{p}_{j,t} = \frac{\kappa_j + 1}{\kappa_j + 2}\left[\frac{A_{j,t+1}^{-1}}{1 - (\kappa_j + 1)l_{j,t}A_{j,t+1}/B_j}\right] \tag{65}$$

Growth Pass-Through. Taking logs of (62) and using $\bar{p}_{j,t} = z_{j,t}/(g_j A_{j,t})$,

$$\log P_{j,t} = \log z_{j,t} - \log g_j - \log A_{j,t} - \log 2 + \log \Lambda_{j,t}. \tag{66}$$

Holding $l_{j,t}$ fixed, g_j is a constant so $\log g_j$ drops out under differentiation. Differentiating (66) with respect to $\log A_{j,t}$

and defining $\rho_{j,t} \equiv -d\log P_{j,t}/d\log A_{j,t}$ gives

$$\rho_{j,t} = 1 - \left(1 + \frac{d\log \Lambda_{j,t}}{d\log z_{j,t}}\right) \frac{d\log z_{j,t}}{d\log A_{j,t}} \tag{67}$$

the formula stated in the main text. Writing $a_j \equiv \kappa_j/(\kappa_j+2)$, $b_j \equiv \kappa_j/(\kappa_j+1)$,

$$\frac{d\log \Lambda_{j,t}}{d\log z_{j,t}} = \frac{2a_j z_{j,t}^{-2}}{1-a_j z_{j,t}^{-2}} - \frac{b_j z_{j,t}^{-1}}{1-b_j z_{j,t}^{-1}} = \frac{-b_j(z_{j,t}-1)(z_{j,t}-a_j)}{z_{j,t}^3(1-a_j z_{j,t}^{-2})(1-b_j z_{j,t}^{-1})} \le 0, \tag{68}$$

$$\frac{d\log z_{j,t}}{d\log A_{j,t}} = \frac{(\kappa_j+1)\, g_j\, l_{j,t}\, A_{j,t}}{B_j - (\kappa_j+1)\, g_j\, l_{j,t}\, A_{j,t}} > 0, \tag{69}$$

positive provided $B_j > (\kappa_j+1) g_j l_{j,t} A_{j,t}$, i.e. in any interior equilibrium. Clearing denominators in (68),

$$1 + \frac{d\log \Lambda_{j,t}}{d\log z_{j,t}} = \frac{z_{j,t}^3 - 2b_j z_{j,t}^2 + a_j z_{j,t}}{z_{j,t}^3 - b_j z_{j,t}^2 - a_j z_{j,t} + a_j b_j} = \frac{z_{j,t}\big[(z_{j,t}-b_j)^2 + (a_j - b_j^2)\big]}{z_{j,t}^3 - b_j z_{j,t}^2 - a_j z_{j,t} + a_j b_j} > 0 \tag{70}$$

Combining this with (69) in (67), growth pass-through is incomplete in any interior equilibrium.

Comparative Statics. Fix $(g_j, l_{j,t}, A_{j,t}, \alpha, \gamma, \chi_j)$ and a value $z_{j,t} \ge 1$. Write

$$\rho_{j,t} = 1 - \underbrace{\left(1 + \frac{d\log \Lambda_{j,t}}{d\log z_{j,t}}\right)}_{\equiv A(\kappa_j, z_{j,t}) > 0} \underbrace{\frac{(\kappa_j+1)\, g_j\, l_{j,t}\, A_{j,t}}{B_j - (\kappa_j+1)\, g_j\, l_{j,t}\, A_{j,t}}}_{\equiv B(\kappa_j) > 0}, \qquad B_j \equiv \frac{\alpha\kappa_j(1-\chi_j)}{2\gamma}. \tag{71}$$

Step 1 (B decreasing in κ_j). Write $B(\kappa_j) = N(\kappa_j)/D(\kappa_j)$ with $N(\kappa_j) = (\kappa_j+1) g_j l_{j,t} A_{j,t}$ and $D(\kappa_j) = B_j - (\kappa_j+1) g_j l_{j,t} A_{j,t}$; both $g_j, l_{j,t}, A_{j,t}, \alpha, \gamma, \chi_j$ are held fixed, so $N'(\kappa_j) = g_j l_{j,t} A_{j,t}$ and $D'(\kappa_j) = (1-\chi_j)\alpha/(2\gamma) - g_j l_{j,t} A_{j,t}$. By the quotient rule,

$$\frac{dB}{d\kappa_j} = \frac{N'D - ND'}{D^2}, \qquad N'D - ND' = g_j l_{j,t} A_{j,t}\big[D - (\kappa_j+1)D'\big].$$

Substituting D and D', the $(\kappa_j+1) g_j l_{j,t} A_{j,t}$ terms cancel exactly, leaving

$$D - (\kappa_j+1)D' = -\frac{(1-\chi_j)\alpha}{2\gamma} < 0 \quad \Longrightarrow \quad N'D - ND' = -\frac{g_j l_{j,t} A_{j,t}(1-\chi_j)\alpha}{2\gamma} < 0.$$

Since $D(\kappa_j) > 0$, $D^2 > 0$, so $dB/d\kappa_j < 0$: $B(\kappa_j)$ is strictly decreasing.

Step 2 (A non-increasing in κ_j). From $\Lambda_{j,t} = (1-a_j z^{-2})/(1-b_j z^{-1})$ with $a_j \equiv \kappa_j/(\kappa_j+2)$, $b_j \equiv \kappa_j/(\kappa_j+1)$, some algebra gives the closed form

$$A(\kappa_j, z) = \frac{z\big[(z-b_j)^2 + (a_j - b_j^2)\big]}{z^3 - b_j z^2 - a_j z + a_j b_j},$$

consistent with the incomplete-pass-through step earlier. Differentiating with respect to κ_j holding z fixed yields

$$\frac{\partial \mathcal{A}(\kappa, z)}{\partial \kappa} = -\frac{z(z-1)\left[\kappa^2(z-1)^3 + 4\kappa z^2(z-1) + 4z^3\right]}{(\kappa z - \kappa + z)^2(\kappa z^2 - \kappa + 2z^2)^2} \leq 0. \tag{72}$$

with strict inequality for $z > 1$.

Step 3. Differentiating (71) with respect to κ_j,

$$\frac{\partial \rho_{j,t}}{\partial \kappa_j} = -\frac{\partial A}{\partial \kappa_j} B - A \frac{\partial B}{\partial \kappa_j}.$$

For $z_{j,t} \geq 1$: $A > 0$, $B > 0$, $\partial B / \partial \kappa_j < 0$ (Step 1), and $\partial A / \partial \kappa_j \leq 0$ (Step 2), so $\partial \rho_{j,t} / \partial \kappa_j > 0$.

Endogenous Markup Growth. Active producers have $a \geq A_{j,t+1}$. Using $p(a) = \frac{1}{2}(\bar{p}_{j,t} + 1/a)$ and $q(a) = \frac{\alpha}{2\gamma}(1 - 1/(a\bar{p}_{j,t}))$, revenue and variable-cost integrands simplify to

$$p(a)q(a) = \frac{\alpha}{4\gamma}\left[\bar{p}_{j,t} - \frac{1}{a^2 \bar{p}_{j,t}}\right], \qquad a^{-1}q(a) = \frac{\alpha}{2\gamma}\left[\frac{1}{a} - \frac{1}{a^2 \bar{p}_{j,t}}\right].$$

The cost-weighted sectoral markup is thus given by:

$$\mu^c_{j,t} = \frac{\int p(a)q(a)\, g_{j,t}(a)\, da}{\int a^{-1}q(a)\, g_{j,t}(a)\, da} = \frac{z_{j,t}\left(1 - \dfrac{\kappa_j}{(\kappa_j + 2)z^2_{j,t}}\right)}{2\kappa_j\left(\dfrac{1}{\kappa_j + 1} - \dfrac{1}{(\kappa_j + 2)z_{j,t}}\right)}.$$

Multiplying numerator and denominator by $(\kappa_j + 2)z_{j,t}$ and simplifying,

$$\mu^c_{j,t} = \frac{\kappa_j + 1}{2\kappa_j} \frac{(\kappa_j + 2)z^2_{j,t} - \kappa_j}{(\kappa_j + 2)z_{j,t} - (\kappa_j + 1)} \tag{73}$$

which is the markup formula stated in the main text. At $z_{j,t} \to 1$ this reduces to $\mu^c_{j,t} \to (\kappa_j + 1)/\kappa_j = 1 + 1/\kappa_j$, recovering the level-shift markup formula of the static (free-entry) model as the special case in which the choke price equals the marginal producer's marginal cost. Over time, as technology progresses, we have increasing markups even with constant κ_j since

$$\frac{\partial \mu^c_{j,t}}{\partial z_{j,t}} = \frac{(\kappa_j + 1)(\kappa_j + 2)(z_{j,t} - 1)\left[\kappa_j(z_{j,t} - 1) + 2z_{j,t}\right]}{2\kappa_j\left[(\kappa_j + 2)z_{j,t} - \kappa_j + 1)\right]^2} \geq 0. \tag{74}$$

B Data: WIOD

B.1 Sector Aggregation

Table B.1: WIOD Rev. 3.1 and Rev. 4 Aggregation into Sectors

3-sector	8-sector	ISIC Rev. 3.1	ISIC Rev. 4
Agriculture	Agriculture	**A** Agriculture **B** Fishing	**A** Agriculture, Forestry & Fishing
Industry	Manufacturing	**D** Manufacturing	**C** Manufacturing **J58** Publishing Activities
	Other Industry	**C** Mining & Quarrying **E** Electricity, Gas & Water **F** Construction	**B** Mining & Quarrying **D** Electricity, Gas, Steam & Air Conditioning Supply **E36** Water Collection & Supply **F** Construction
Services	Trade Services	**G** Wholesale & Retail Trade **H** Accommodation and Food	**G** Wholesale & Retail Trade **I** Accommodation and Food
	Transport & Communication	**I** Transport & Storage **I64** Post & Telecom	**H** Transport, Storage and Post **J61** Telecom
	Finance	**J** Financial and Insurance	**K** Financial and Insurance
	Business Services	**K** Real Estate, Renting & Business Activities	**L68** Real Estate **M** Professional, Scientific & Technical Activities **N** Administrative & Support Service Activities
	Other Services	**L–Q** Public Administration and Defence; Education; Health and Social Work; Other Community, Social and Personal Service Activities; Private Households; Extraterritorial Organizations	**E37–E39** Sewerage, Waste management and remediation **J59–J60** Motion picture, sound recording, broadcasting **J62–J63** Computer programming, consultancy, information Services **O–U** Public Administration and Defence; Education; Health and Social Work; Arts, Entertainment and Recreation; Other Service Activities; Private Households; Extraterritorial Organizations

B.2 Expenditure Shares

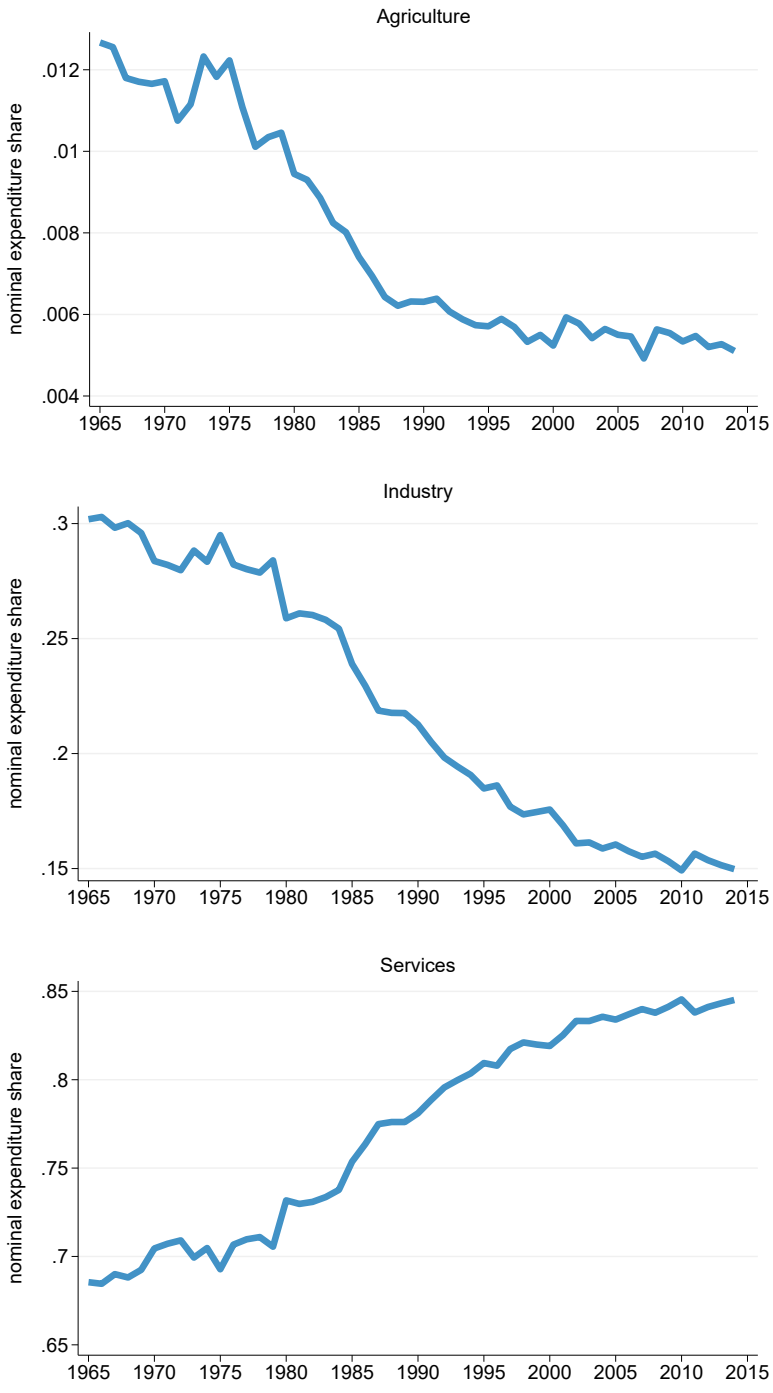
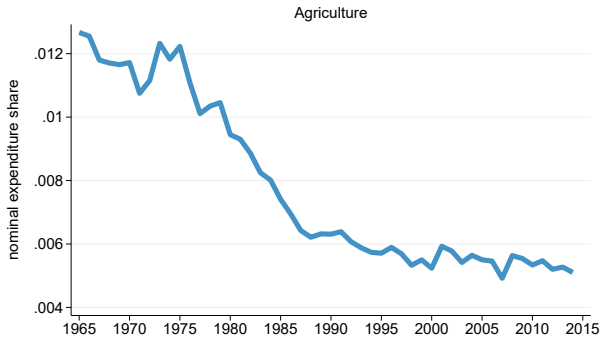
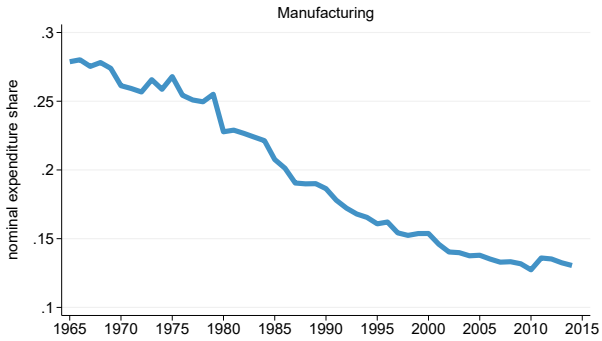
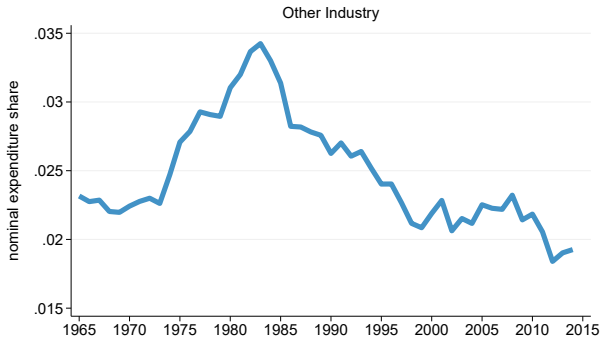


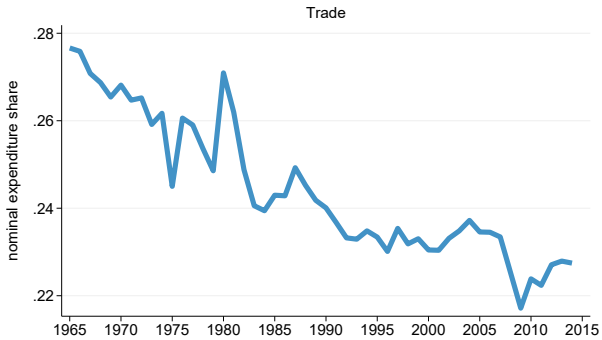
Figure B.1: Expenditure Shares across 3 Sectors

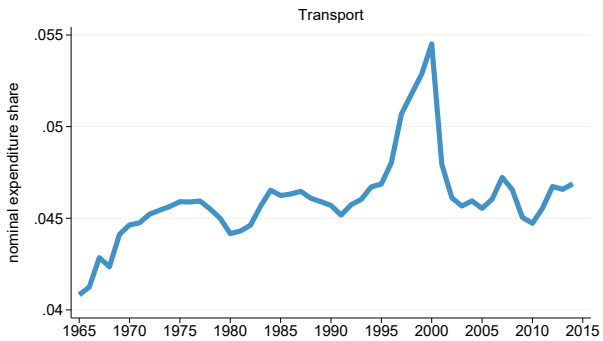
Notes: Own calculations based on WIOD. Expenditure shares are constructed using final household consumption and then aggregated into 3 broad economic sectors: Agriculture, Industry, and Services.

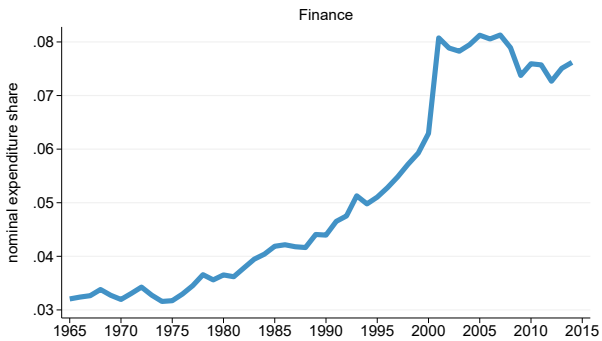

Agriculture
nominal expenditure share
.004
.006
.008
.01
.012
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015


Manufacturing
nominal expenditure share
.1
.15
.2
.25
.3
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015


Other Industry
nominal expenditure share
.015
.02
.025
.03
.035
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015


Trade
nominal expenditure share
.22
.24
.26
.28
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015


Transport
nominal expenditure share
.04
.045
.05
.055
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015


Finance
nominal expenditure share
.03
.04
.05
.06
.07
.08
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015

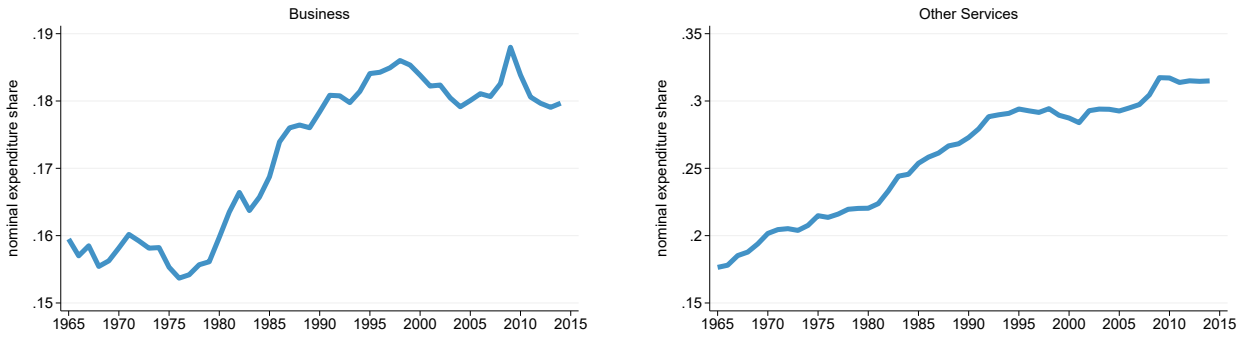


Figure B.2: Expenditure Shares across 8 Sectors

Notes: Own calculations based on WIOD. Expenditure shares are constructed using final household consumption and then aggregated into 8 sectors: Agriculture, Manufacturing, Other Industry, Trade Services, Transport Services, Finance, Business Services, and Other Services.

B.3 Relative Prices

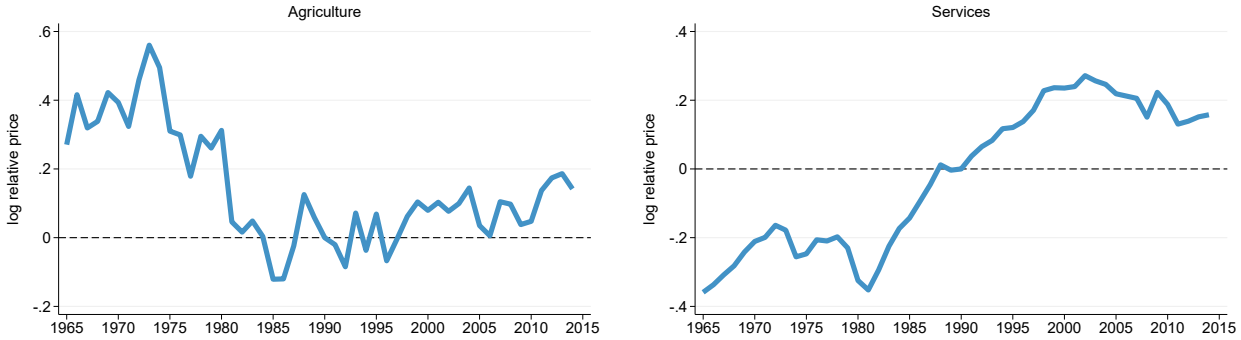


Figure B.3: Sectoral Prices relative to Industry (3-sectors)

Notes: Own calculations based on WIOD. The figure plots sectoral relative price levels for Agriculture and Services relative to Industry. Sectoral gross price changes are measured using the implicit deflator $g^P_{j,t} \equiv E^{\text{nom}}_{j,t}/E^{\text{pyp}}_{j,t}$ (nominal expenditure divided by previous-year-price expenditure). A chained log price index is constructed as $\log P_{j,t} - \log P_{j,t_0} = \sum_{\tau=t_0+1}^{t} \log(g^P_{j,\tau})$. Displayed is the log relative price index, $\log(P_{j,t}/P_{b,t}) - \log(P_{j,t_0}/P_{b,t_0})$, with normalization year $t_0 = 1990$.

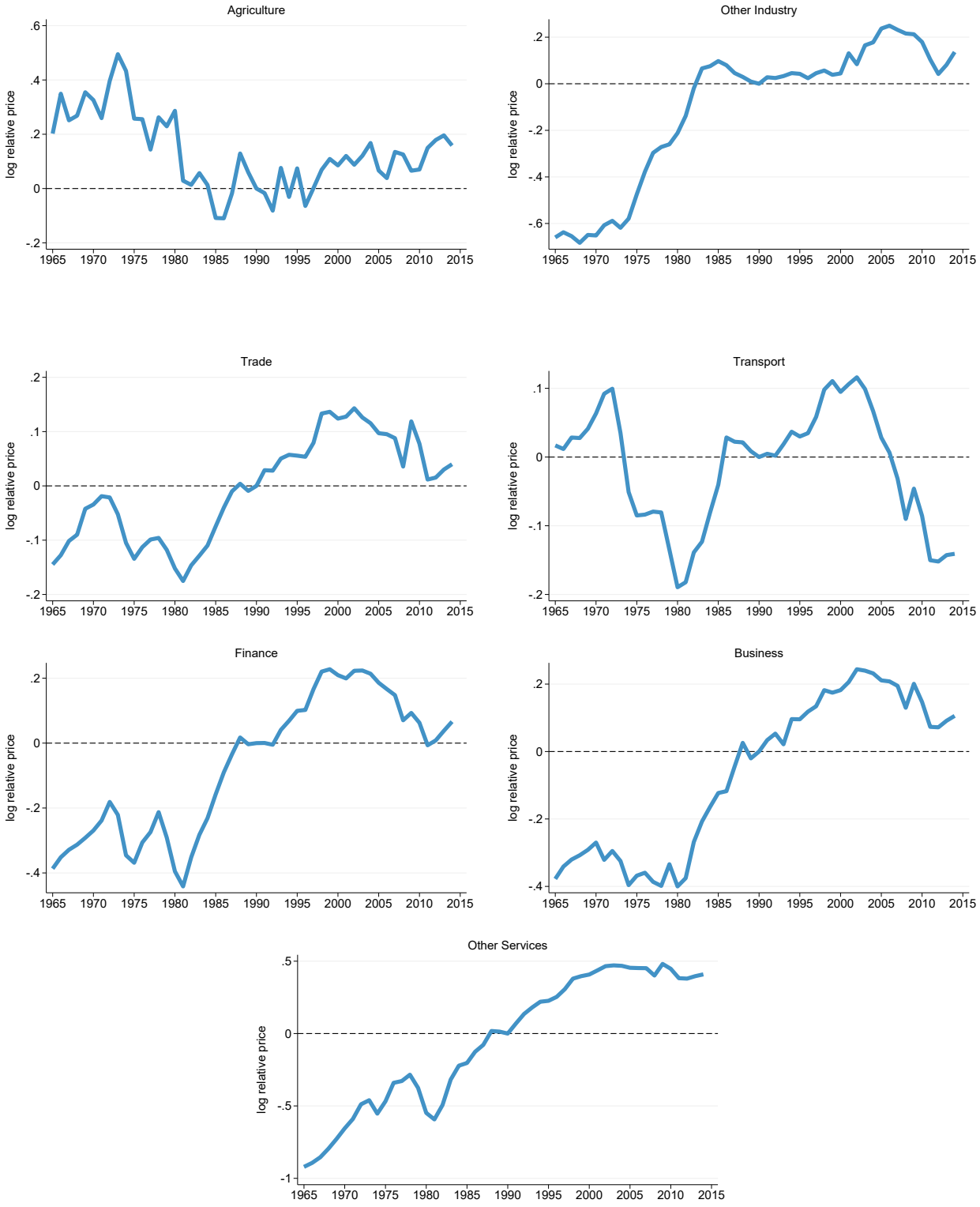


Figure B.4: Sectoral Prices relative to Manufacturing (8-sectors)

Notes: Own calculations based on WIOD. The figure plots sectoral relative price levels for all other sectors relative to Manufacturing. Sectoral gross price changes are measured using the implicit deflator $g^P_{j,t} \equiv E^{\text{nom}}_{j,t}/E^{\text{pyp}}_{j,t}$ (nominal expenditure divided by previous-year-price expenditure). A chained log price index is constructed as $\log P_{j,t} - \log P_{j,t_0} = \sum_{\tau=t_0+1}^{t} \log(g^P_{j,\tau})$. Displayed is the log relative price index, $\log(P_{j,t}/P_{b,t}) - \log(P_{j,t_0}/P_{b,t_0})$, with normalization year $t_0 = 1990$.

B.4 GMM Estimation Details

I estimate the cross-sector demand parameters using the differenced relative expenditure share equation (27). Since the aggregate price index is not directly observed, I recover changes in aggregate real consumption using the expenditure share of the base sector b:

$$\Delta \log Q_t = \Delta \log E_t - \Delta \log P_{b,t} + \frac{1}{1-\eta} \Delta \log \omega_{b,t}. \tag{75}$$

Substituting this expression into (27) yields the nonlinear estimating equation

$$\begin{aligned} \Delta \log \left(\frac{\omega_{j,t}}{\omega_{b,t}} \right) = (1-\eta) \Delta \log \left(\frac{P_{j,t}}{P_{b,t}} \right) \\ + (\epsilon_j - 1) \left[\Delta \log E_t - \Delta \log P_{b,t} + \frac{\Delta \log \omega_{b,t}}{1-\eta} \right] + u_{j,t}, \end{aligned} \tag{76}$$

where $u_{j,t}$ denotes the estimation residual and E_t is aggregate nominal household expenditure per capita.

For each non-base sector j, define the vector of instruments

$$Z_{j,t} = \begin{pmatrix} 1 \\ \Delta \log(P_{j,t}/P_{b,t}) \\ \Delta \log E_t - \Delta \log P_{b,t} \\ \Delta \log \omega_{b,t} \end{pmatrix}. \tag{77}$$

The GMM estimator is based on the following moment conditions, which impose orthogonality between the instruments and the estimation residual:

$$\mathbb{E}\left[Z_{j,t} u_{j,t} \right] = 0, \qquad j \in \mathcal{J} \setminus \{b\}. \tag{78}$$

The instruments consist of the contemporaneous variables entering the estimating equation and a constant; no additional excluded instruments are used. I jointly estimate the equations for all non-base sectors using one-step non-linear GMM with an identity weighting matrix.

To impose the theoretical restriction $0 < \eta < 1$, I parametrise $1 - \eta$ using the logistic transformation

$$1 - \eta = \frac{\exp(\theta)}{1 + \exp(\theta)}, \tag{79}$$

where θ is unrestricted. The relative income elasticities $\epsilon_j - 1$ are estimated directly, with the income elasticity of the base sector normalized to $\epsilon_b = 1$. The reported elasticity of substitution is recovered by inverting the logistic transformation. Confidence intervals for the transformed parameters are calculated using the delta method.

The estimation uses annual US household consumption data from WIOD over 1965–2014. First differencing leaves 1966–2014 as the estimation period. For the three-sector specification, Industry serves as the base sector, yielding 98 sector-year observations. For the eight-sector specification, Manufacturing is the base sector. Due to discontinuities between the WIOD releases in the more disaggregated data, I exclude the 2001 first-difference observations in the eight-sector specification, leaving 336 observations. All specifications are estimated separately using domestic and trade-inclusive expenditure shares and price changes. Time-invariant sector-specific demand shifters difference out and therefore do not require additional sector fixed effects.

C Data: Compustat

C.1 Sector Aggregation

Table C.1: Compustat NAICS 2-Digit Aggregation into 3 Sectors

3-sector	**NAICS 2-Digit Industry**	
Agriculture	11	Agriculture
Industry	21	Mining
	22	Utilities
	23	Construction
	31	Manufacturing non-durables
	32	Manufacturing durables
	33	Manufacturing misc
Services	42	Wholesale Trade
	44	Retail Trade
	45	Retail Trade
	48	Transport, Warehouse
	51	Information
	52	Finance and Insurance
	53	Real Estate
	54	Professional Services
	56	Administrative Support
	62	Education, Health Care
	71	Arts and Recreation
	72	Accommodation and Food
	81	Other Services

C.2 Derivation Markup Estimation Equation

To see where equation (28) comes from, consider a firm that minimizes variable and fixed input costs subject to producing Q_{it}:

$$\min_{V_{it},K_{it}} P_{it}^V V_{it} + R_{it}K_{it} \quad \text{j.t.} \quad Q_{it} \leq F(V_{it}, K_{it}, \Omega_{it}). \tag{80}$$

Let λ_{it} denote the Lagrange multiplier on the production constraint. This multiplier is marginal cost. The first-order condition for the variable input is

$$P_{it}^V = \lambda_{it} \frac{\partial F(\cdot)}{\partial V_{it}}. \tag{81}$$

Multiplying both sides by V_{it}/Q_{it} gives

$$\frac{\partial F(\cdot)}{\partial V_{it}} \frac{V_{it}}{Q_{it}} = \frac{P_{it}^V V_{it}}{\lambda_{it} Q_{it}}. \tag{82}$$

The left-hand side is the output elasticity of the variable input, denoted θ_{it}^V. Using the definition of the markup, $\mu_{it} = P_{it}/\lambda_{it}$, we obtain

$$\theta_{it}^V = \frac{P_{it}^V V_{it}}{P_{it} Q_{it}} \mu_{it}. \tag{83}$$

Solving for the markup yields

$$\mu_{it} = \theta_{it}^V \frac{P_{it} Q_{it}}{P_{it}^V V_{it}} = \theta_{it}^V \frac{R_{it}}{C_{it}}. \tag{84}$$

C.3 Aggregate Markup

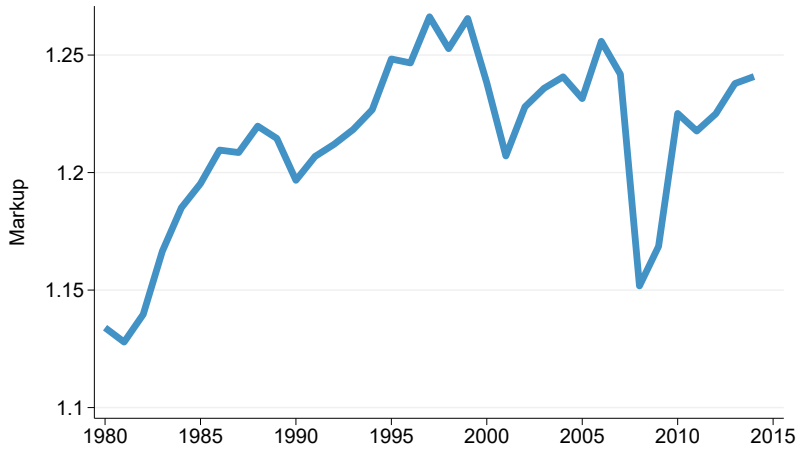
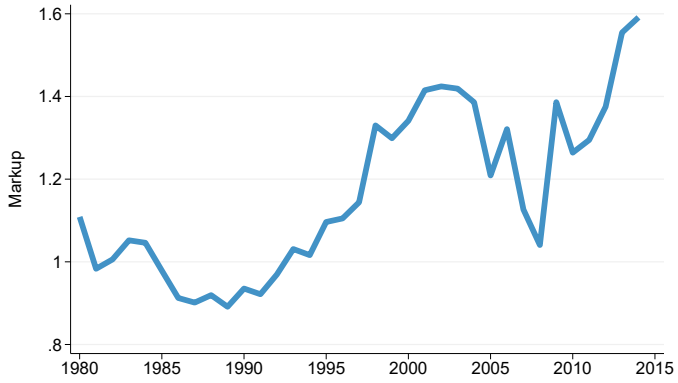


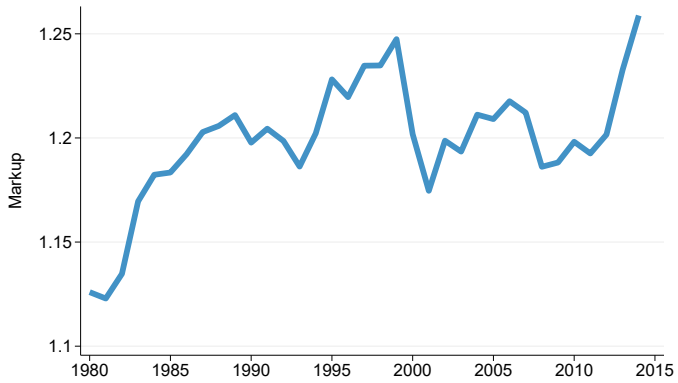
Figure C.1: Aggregate Markup

Notes: The figure shows the aggregate cost-weighted markup for the cleaned Compustat firm-year sample after applying the 1% trimming procedure described in the text. Firm-level markups are constructed as the product of the output elasticity of the variable input and the inverse COGS share of sales. The aggregate markup is computed as the COGS-weighted average.

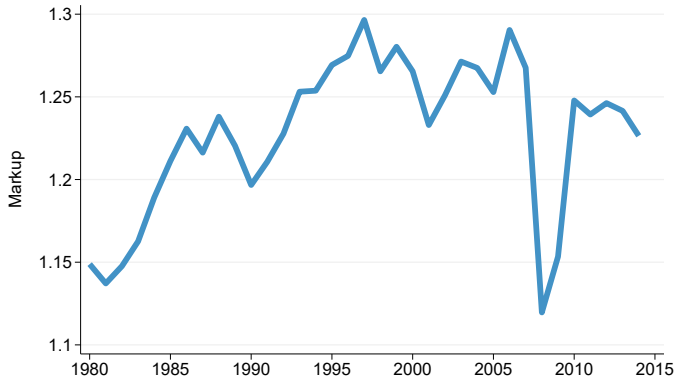
C.4 Sectoral Markups



(a) Agriculture



(b) Industry



(c) Services

Figure C.2: Sectoral Aggregate Markups

Notes: The figures show the aggregate cost-weighted markup within each sector for the cleaned Compustat firm-year sample after applying the 1% trimming procedure described in the text. Firm-level markups are constructed as the product of the output elasticity of the variable input and the inverse COGS share of sales. The sectoral aggregate markup is computed as the COGS-weighted average of all firms in the sector.

C.5 Decomposition of Aggregate Markup Change

Table C.2: Decomposition of Aggregate Markup Change

	COGS share		Contribution to ΔM		
Sector	**1980**	**2014**	**Markup**	**Share**	**Total**
Agriculture	0.4%	0.1%	0.001	-0.004	-0.003
Industry	64.2%	42.6%	0.071	-0.258	-0.187
Services	35.4%	57.3%	0.036	0.261	0.297
Total change			0.108	-0.001	0.107

Notes: The left part reports each sector's share in aggregate COGS, which is used as the aggregation weight for sectoral markups. The right part decomposes the change in the aggregate markup between 1980 and 2014 into two components: a within-sector markup effect, capturing changes in average sectoral markups holding sector size fixed, and a sector-size effect, capturing changes in sectoral COGS shares holding markups fixed. Sector markups are COGS-weighted averages of firm-level markups. The decomposition is $\Delta\mathcal{M} = \sum_{j\in\mathcal{J}} \bar{\omega}_j^c \Delta\mu_j^c + \sum_{j\in\mathcal{J}} \bar{\mu}_j^c \Delta\omega_j^c$, where μ_j^c denotes the sectoral markup, ω_j^c denotes the sectoral COGS share, and bars denote averages of initial and final values.

D Calibration & Counterfactual

D.1 Productivity Growth Decomposition

Table D.1: Productivity Growth Decomposition: Domestic

Sector	**Cut-off a^***	**Distribution**	**Total**
Agriculture	2.0%	2.3%	4.3%
Industry	1.9%	0.5%	2.4%
Services	0.6%	0.3%	0.9%

Notes: Values are average annual log changes. Conditional-mean productivity among producing firms, $E[a_{j,t} \mid a_{j,t} \geq a_{j,t}^*] = a_{j,t}^* \cdot \kappa_{j,t}/(\kappa_{j,t} - 1)$, is decomposed into $\Delta \log E[a_{j,t} \mid a_{j,t} \geq a_{j,t}^*] = \Delta \log a_{j,t}^* + \Delta \log[\kappa_{j,t}/(\kappa_{j,t} - 1)]$. The cut-off effect captures the shift in the productivity threshold of producing firms; the distribution effect captures the change in the shape of the productivity distribution above that threshold. Since this only depends on $\kappa_{j,t}$, it is identical across the domestic and trade-inclusive specifications. The baseline uses the smoothed series for κ.

D.2 Price Index Decomposition

Table D.2: Decomposition of Price Growth: Trade-inclusive

Sector	**Cut-off**	**Selection Effect**	**Price**
Agriculture	2.67%	0.65%	3.32%
Industry	2.39%	0.24%	2.62%
Services	0.62%	0.14%	0.76%

Notes: Values are average annual log changes. Wage-adjusted price growth is decomposed according to $\Delta \log(P_{j,t}/w_t)^{-1} = \Delta \log a_{j,t}^* + \Delta \log[(\kappa_{j,t} + 2)/(\kappa_{j,t} + 1)]$. The selection effect can equivalently be written as $\Delta \log[(2\mu_{j,t}^c - 1)/\mu_{j,t}^c]$.

D.3 Prices & Productivity



(a) Agriculture (b) Industry (c) Services

Figure D.1: TFP and Price Indices

Notes: The solid line plots the constructed TFP index, while the dashed line plots the wage-adjusted price index, $P_{j,t}/w_t$. Both series are normalized to 100 in the first year. The discrepancies between TFP growth and price growth reflect markup growth.

D.4 Sensitivity to K

Table D.3: Sensitivity to K

	2014 share gap (in pp)		
K	**Agriculture**	**Industry**	**Services**
-2.0	-0.050	-1.218	1.268
-1.0	-0.163	-1.955	2.118
0.0	-0.255	-2.667	2.921
1.0	-0.329	-3.352	3.681
2.0	-0.389	-4.013	4.402

Notes: full-pass-through counterfactual minus baseline expenditure share in 2014. K enters through the structural recovery of $A_{j,t}$ from $a^*_{j,t}$.

D.5 Real Consumption

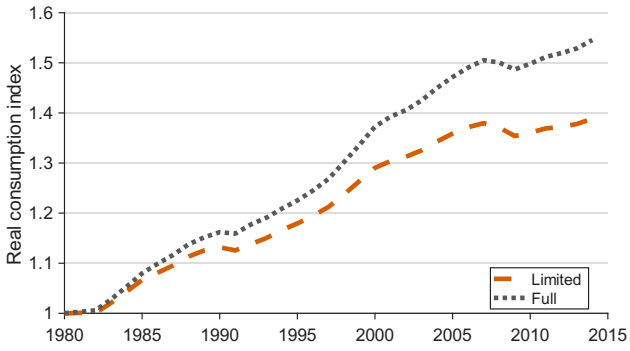
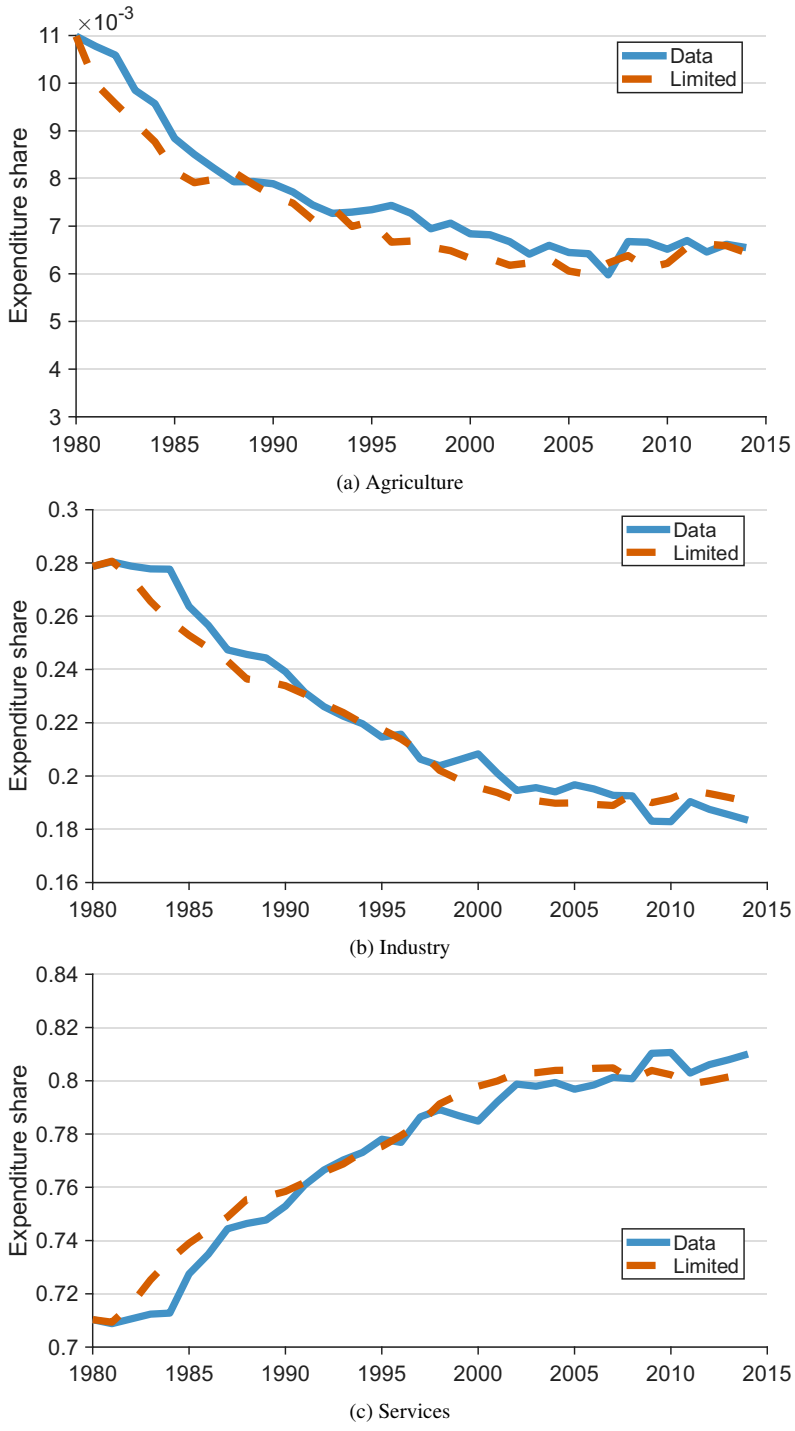


Figure D.2: Real Consumption Index

Notes: The figure plots aggregate real consumption under the limited pass-through baseline and the full pass-through counterfactual, normalized to one in 1980. Changes in aggregate real consumption generate the income-effect in the changing expenditure shares.

D.6 Baseline Calibration



(a) Agriculture

(b) Industry

(c) Services

Figure D.3: Calibrated Expenditure Shares

Notes: Each panel compares the observed expenditure share with the model-implied share under the limited pass-through baseline. By construction, the baseline matches the observed expenditure shares exactly in 1980.

E Extensions

E.1 8-Sector Aggregation

Table E.1: Cross Sector Parameter Estimates: 8-Sector Aggregation

	(1)	(2)
η	0.11	0.51
	[-0.14,0.36]	[0.05,0.97]
Agriculture	-0.36	-0.13
	[-0.59,-0.13]	[-0.42,0.16]
Other Industry	-0.23	-0.27
	[-0.40,-0.05]	[-0.51,-0.03]
Trade	0.48	0.08
	[0.19,0.76]	[-0.24,0.40]
Transport	1.55	0.90
	[0.88,2.21]	[-0.11,1.91]
Finance	1.23	0.93
	[0.48,1.99]	[-0.49,2.36]
Business	0.57	0.31
	[0.13,1.01]	[-0.43,1.05]
Other Services	0.12	0.34
	[-0.47,0.71]	[-0.92,1.60]
Sector FE	N	N
Trade Controls	N	Y
Observations	336	336

Notes: The table reports non-linear GMM estimates of the differenced relative expenditure-share equation. Column (1) uses domestic expenditure shares and domestic implicit deflators only. Column (2) uses trade-inclusive expenditure shares and implicit deflators, i.e. imports are included in household expenditure and price measurement. The base sector is manufacturing, so $\epsilon_b = 1$ by normalisation; hence the reported sector coefficients are estimates of $\epsilon_j - 1$ relative to the base sector. Brackets report 95% confidence intervals. Observations are sector-year cells. Because the estimating equation is written in first differences, time-invariant sector fixed effects difference out mechanically; therefore no sector fixed effects are included.

Table E.2: Productivity and Pass-Through: 8-Sector Aggregation

Sector	**Cut-off** a^*	ρ_{1980}	ρ_{2014}
Agriculture	2.7%	90%	63%
Manufacturing	2.5%	87%	77%
Other Industry	1.1%	96%	87%
Trade	1.7%	90%	90%
Transport	2.0%	92%	85%
Finance	0.5%	88%	70%
Business	0.7%	85%	81%
Other Services	-0.4%	70%	74%

Notes: Productivity growth is the average annual log change of $a^*_{j,t}$, the productivity cut-off of producing firms, using the trade-inclusive specification. Cost-weighted markups imply $\kappa_{j,t} = 1/(\mu^c_{j,t} - 1)$. The baseline uses the smoothed kappa series.

E.2 PCE Price Data

Table E.3: Productivity Cut-off: WIOD vs. PCE Prices

Sector	WIOD	PCE	Difference
Agriculture	2.67%	0.73%	-1.94pp
Industry	2.39%	2.61%	0.22pp
Services	0.62%	0.26%	-0.36pp

Notes: Values are average annual log changes, 1980–2014. Both columns are computed as $\Delta \log a^*_{j,t} = \Delta \log[(\kappa_{j,t}+1)/(\kappa_{j,t}+2)] - \Delta \log P_{j,t} + \Delta \log w_t$, using the identical smoothed $\kappa_{j,t}$ path and BLS COMPNFB wage series in both columns; only the price source $P_{j,t}$ differs. WIOD uses the trade-inclusive (baseline) specification. PCE maps Agriculture to PCE food, Services to PCE services, and Industry to PCE goods excluding food. Difference is PCE minus WIOD, in percentage points.